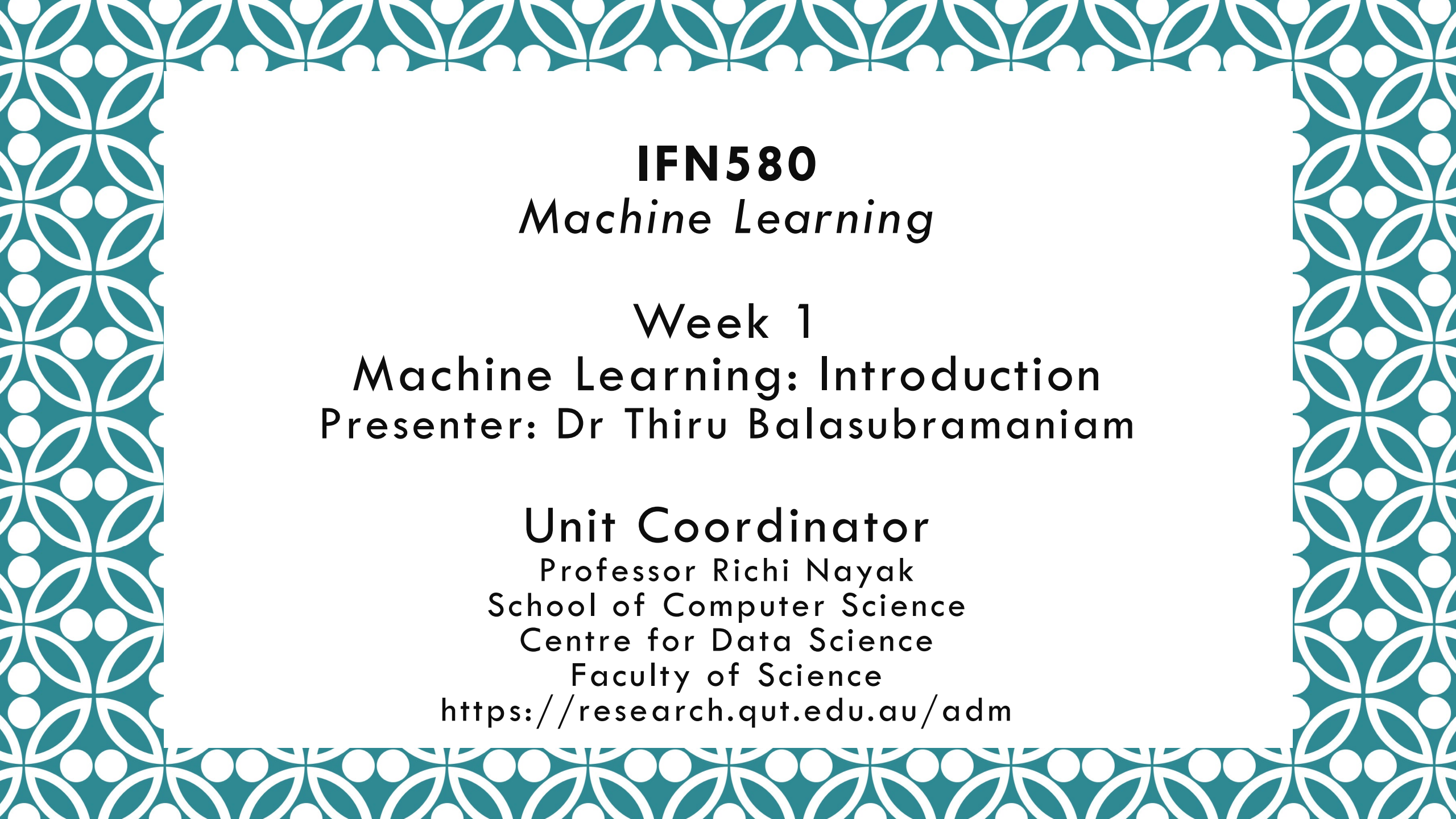


The slide features a decorative border with a repeating geometric pattern of teal leaves and white circles. The central content is on a white background.

IFN580

Machine Learning

Week 1

Machine Learning: Introduction
Presenter: Dr Thiru Balasubramaniam

Unit Coordinator

Professor Richi Nayak
School of Computer Science
Centre for Data Science
Faculty of Science

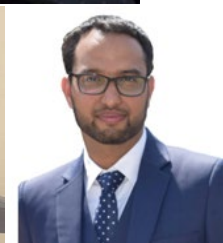
<https://research.qut.edu.au/adm>

<https://research.qut.edu.au/adm/people/richi-nayak/>

Contact: r.nayak@qut.edu.au

<https://research.qut.edu.au/adm/>

Publications: http://eprints.qut.edu.au/view/person/Nayak_Richi.html





Week 1 Learning

- **Lecture – Starting from Week 1**
 - Introduction to the Unit
 - What is Machine Learning?
 - Difference between related areas – ML, AI, DS, Big Data, GenAI
 - Types of Machine Learning
 - Machine Learning Pipeline
- **Supervised Tutorial & Practical – Starting from Week 2**
 - Tutorial (Part 1) - Reflective Pen-and-Paper exercises
 - Supervised vs Unsupervised ML
 - ML Pipeline
 - Data formats: Structured vs Unstructured
 - Practical (Part 2) - Practical Exercises
 - Working with Python
 - Installing Anaconda
 - Working with Jupyter notebook
 - Simple commands
 - Saving the Jupyter notebook file

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IFN580

Unit Details

TEACHING TEAM

Professor Richi Nayak

- Unit Coordinator
- Lecturer

r.nayak@qut.edu.au



Professor Yuefeng Li

- Lecturer

y2.li@qut.edu.au



Dr Thiru Balasubramaniam

- Tutor

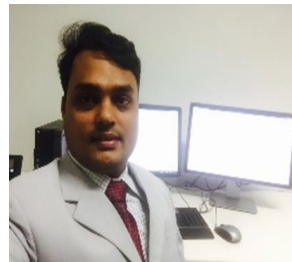
thirunavukarasu.balas@qut.edu.au



Dr Md Abul Bashar

- Tutor

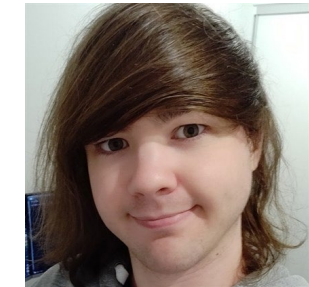
m1.bashar@qut.edu.au



Dr Connor McLaughlin

- Tutor

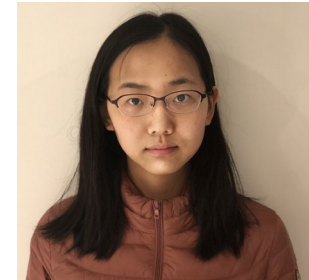
c.mclaughlin@qut.edu.au



Dr Duoyi Zhang

- Tutor

d25.zhang@qut.edu.au



Dr Guangxin Su (Neil)

- Tutor

guangxin.su@qut.edu.au



Communication

- Any formal announcements will be made via Canvas.
- Our informal method of communication is Slack.

You can access our workspace at <https://ifn580-semester1-2026.slack.com>

The workspace sign-up page is:

<https://join.slack.com/t/ifn580-semester1-2026/signup>

You need to use your @connect.qut.edu.au or @qut.edu.au address to register.

Slack is widely used in the industry.

It encourages peer support and allows you to help each other. It also allows us to provide replies effectively (as we do not need to answer the same question multiple times). Your queries will be responded to within 48 hrs.

- Our formal method of communication will still be email. If you have a particular question to be asked, write an email to the teaching team.



What will we learn in IFN580?

- Machine Learning: Introduction
- Data Representation: Data to Features
- Supervised Machine Learning
 - Decision Trees
 - Regression Modelling
 - Neural Networks
- Unsupervised Machine Learning
 - Dimensionality Reduction - PCA
 - Clustering algorithms
- Advanced Machine Learning Concepts
 - Deep Learning
 - Generative AI
 - Large language models

We plan to organise guest lectures to show you how these skills are used in the real world.



How will we learn?

- Lectures
 - F2F/Online lecture, Tuesday 3-5 pm in GP S410.
 - This is where we will cover all the theoretical content.
 - Recording will be available after the lecture is held.
- Tutorials
 - This is where you get the chance to try things by yourself.
 - Tutorial will contain pen and paper exercises.
- Practicals
 - This is where you get the chance to try things by yourself.
 - Practical will contain computer-based tasks with Python (next slide).
- Crucially, we will learn by being **active participants**.
 - Ask questions. All questions are useful.
 - Attempt the exercises in tutorials/practicals.
 - Be respectful and supportive.

Tutorial

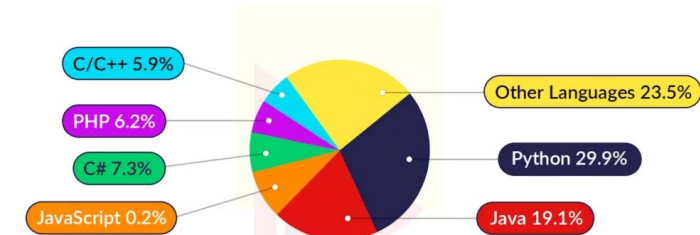
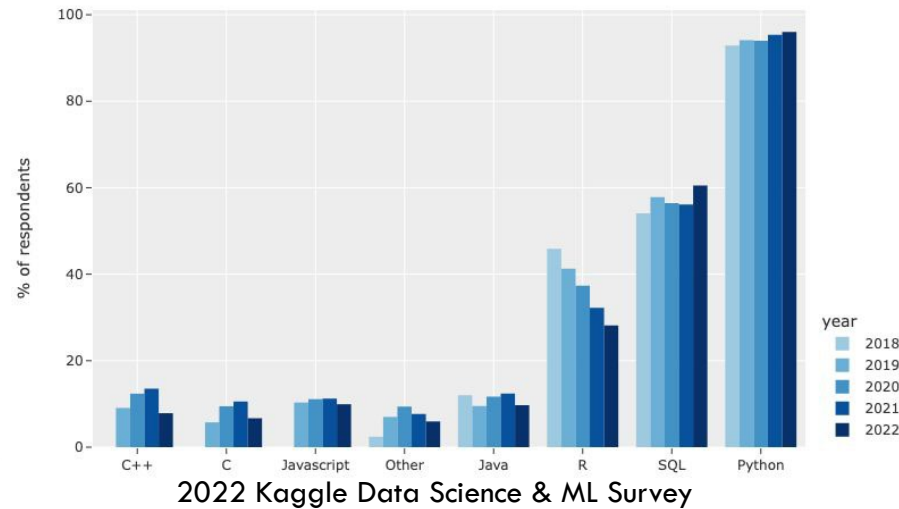
Day	Time	Location
Tuesday	5 – 6 pm	GP S307
Wednesday	2 – 3 pm	GP S309
Wednesday	3 – 4 pm	Online
Wednesday	4 – 5 pm	GP P413
Wednesday	5 – 6 pm	GP B224
Wednesday	6 – 7 pm	GP P504
Wednesday	7 – 8 pm	GP O520

Practical

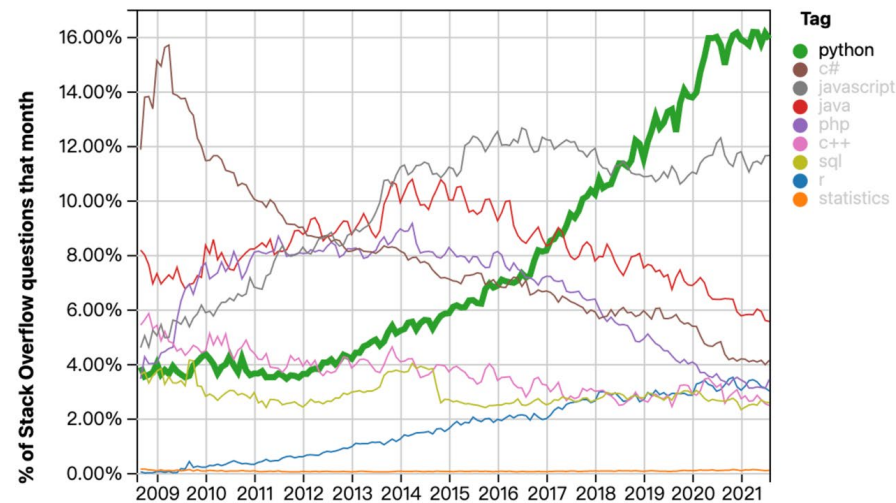
Day	Time	Location
Thursday	1 – 2 pm	GP D416
Thursday	2 – 3 pm	GP D416
Thursday	7 – 8 pm	GP D416
Friday	12 – 1 pm	GP D413
Friday	12 – 1 pm	GP F101
Friday	1 – 2 pm	Online
Friday	1 – 2 pm	GP S502

Why Python?

Most Popular Programming Languages 2018-2022



<https://technobrain.io/watch-out-for-python-trends-2023/>



<https://insights.stackoverflow.com/trends>

- In 2019, over 8.2 million developers used Python.
- Don't need to write much (there are over 137,000 Python libraries).
- Python libraries do the heavy lifting.
- pandas for loading and playing around with data, transforming data inputs into a numerical matrix
- matplotlib for visualizing the data,
- Scikit-learn (or sklearn) for the actual machine learning and assessment.
- Pytorch for deep learning algorithms



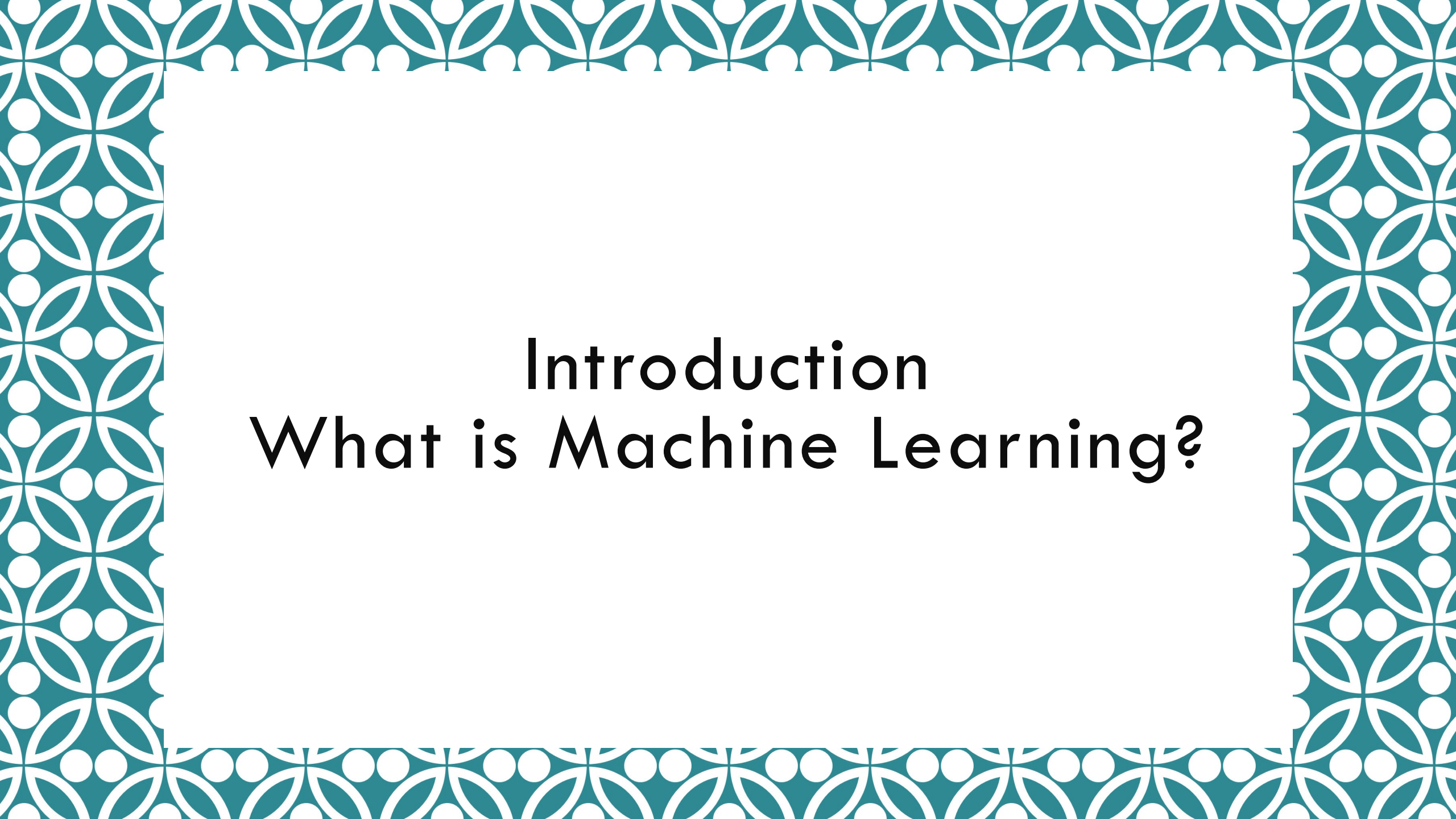
How do we assess?

- Assignment 1: Problem Solving Task
 - This will build directly upon the exercises covered in the computer labs related to supervised ML topics.
 - You will be required to design and develop solutions by applying supervised ML techniques to solve the given data problems.
 - This includes data modelling, interpretation and communication of results, along with critical reflection regarding the applied methods.
 - **It is due in week 7, and it is worth 20 marks.**
- Project (Applied)
 - This will build upon the exercises covered in the computer labs related to unsupervised and advanced ML topics.
 - You will be required to design and develop solutions by applying unsupervised and advanced ML techniques to solve the given data problems.
 - This includes data modelling, interpretation and communication of results along with critical reflection regarding the applied methods.
 - **It is due in week 13, with an online quiz in week 14, and it is worth 40 marks.**
- For Group Work Assignments
 - A team of 3 is an ideal size and only allowed.
 - Not an individual work.
 - Utilise week 2 tutorial/practical to form a team
- Final exam
 - This will be a mix of quiz and short-answer questions.
 - It will take place **during the central examination period** at the end of the semester in-person mode.
- **This subject requires your commitment and efforts.**



Academic integrity and plagiarism

- We take the academic integrity and plagiarism very seriously in this unit.
- **You are NOT allowed to consult with any other groups and/or take help outside (including AI systems such as chatGPT) for assessment tasks.**
- **There may be serious consequences.**
- It is best to take the best advantage of the teaching team, your group members and learn the concepts taught in the unit honestly.

A decorative border with a repeating geometric pattern of teal leaves and white circles surrounds the central text area.

Introduction

What is Machine Learning?



What is Learning?

“The activity or process of gaining knowledge or skill by instruction or study.”
Merriam Webster dictionary

“Learning is any process by which a system improves performance from experience.”
Herbert Simon (1988)

“A computer program is said to learn from *experience E* with respect to some class of *tasks T* and *performance measure P*, if its performance at tasks in *T*, as measured by *P*, improves with experience *E*. -

Definition by Tom Mitchell (1998)

A well-defined learning task can be given by $\langle P, T, E \rangle$.

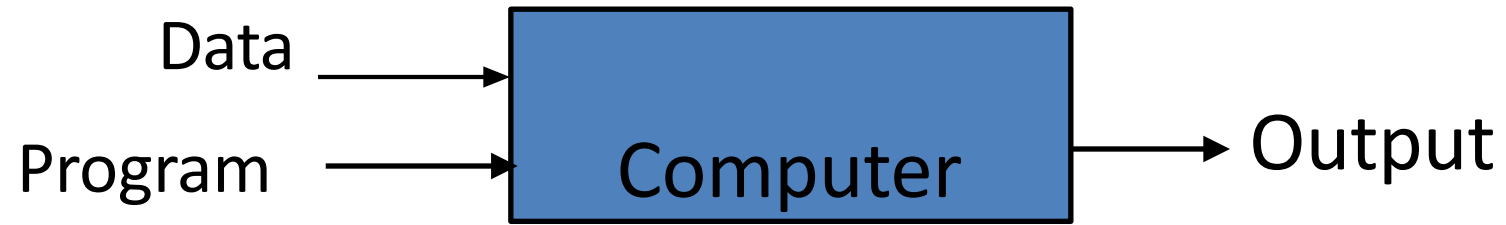
SAMUEL'S CHECKERS-PLAYER

“Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.” Arthur Samuel (1959)

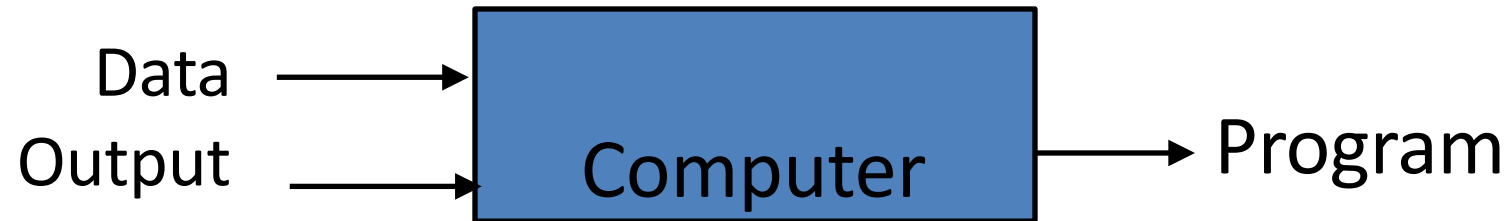




Traditional Programming

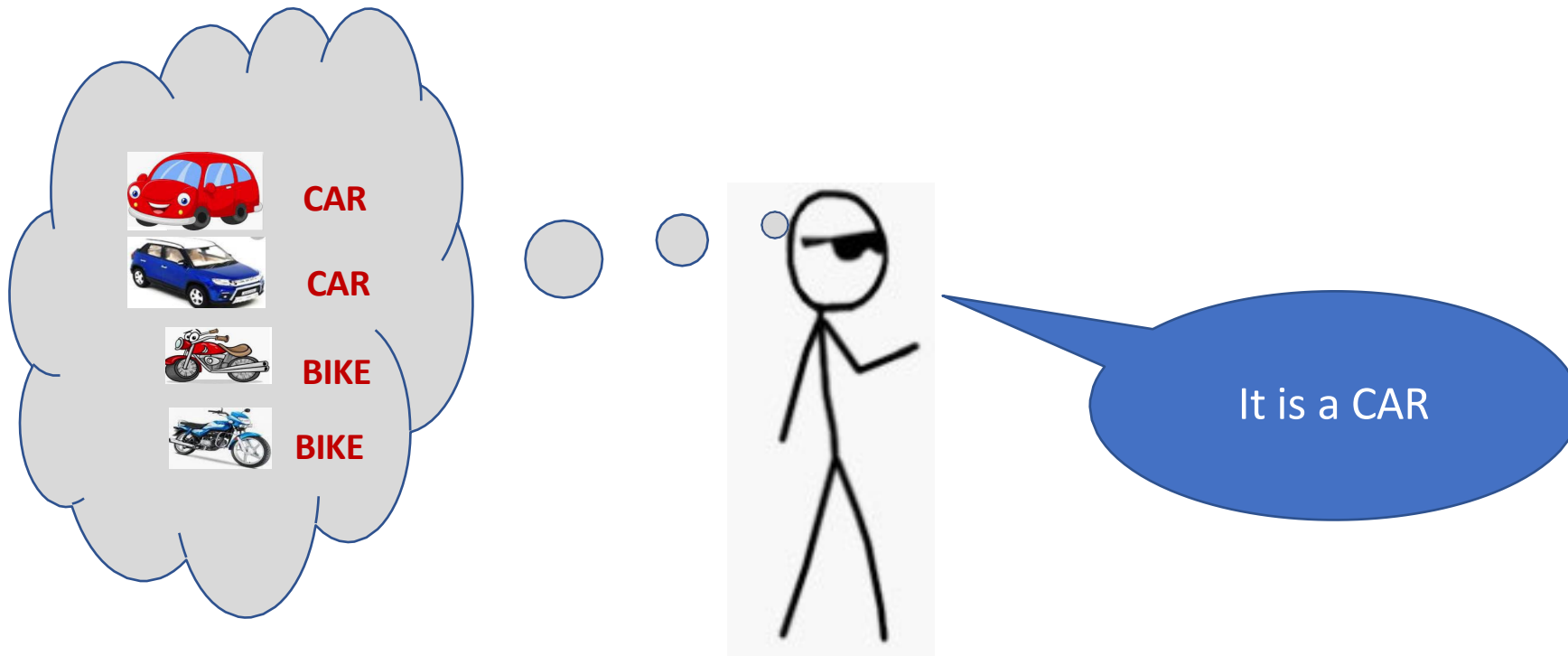


Machine Learning





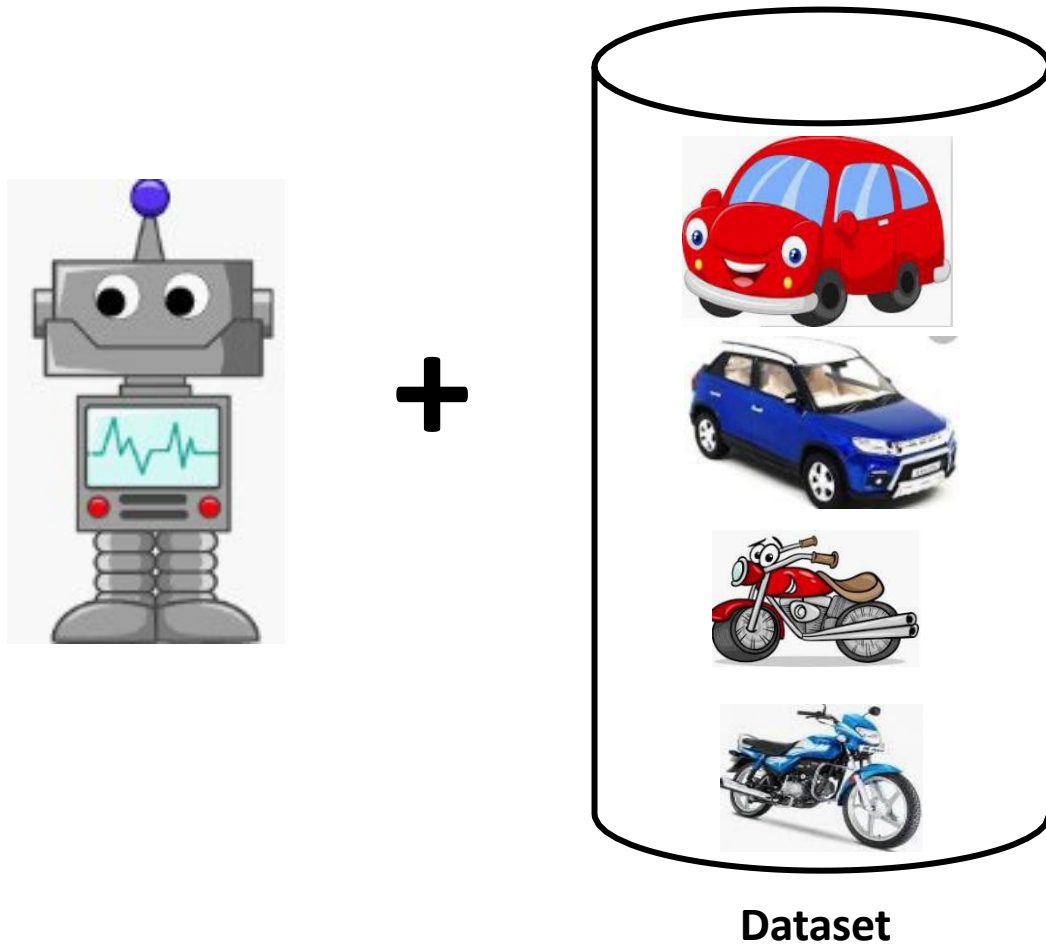
What is this object?



A human can learn from past experiences and make decisions of their own.



What is Machine Learning?

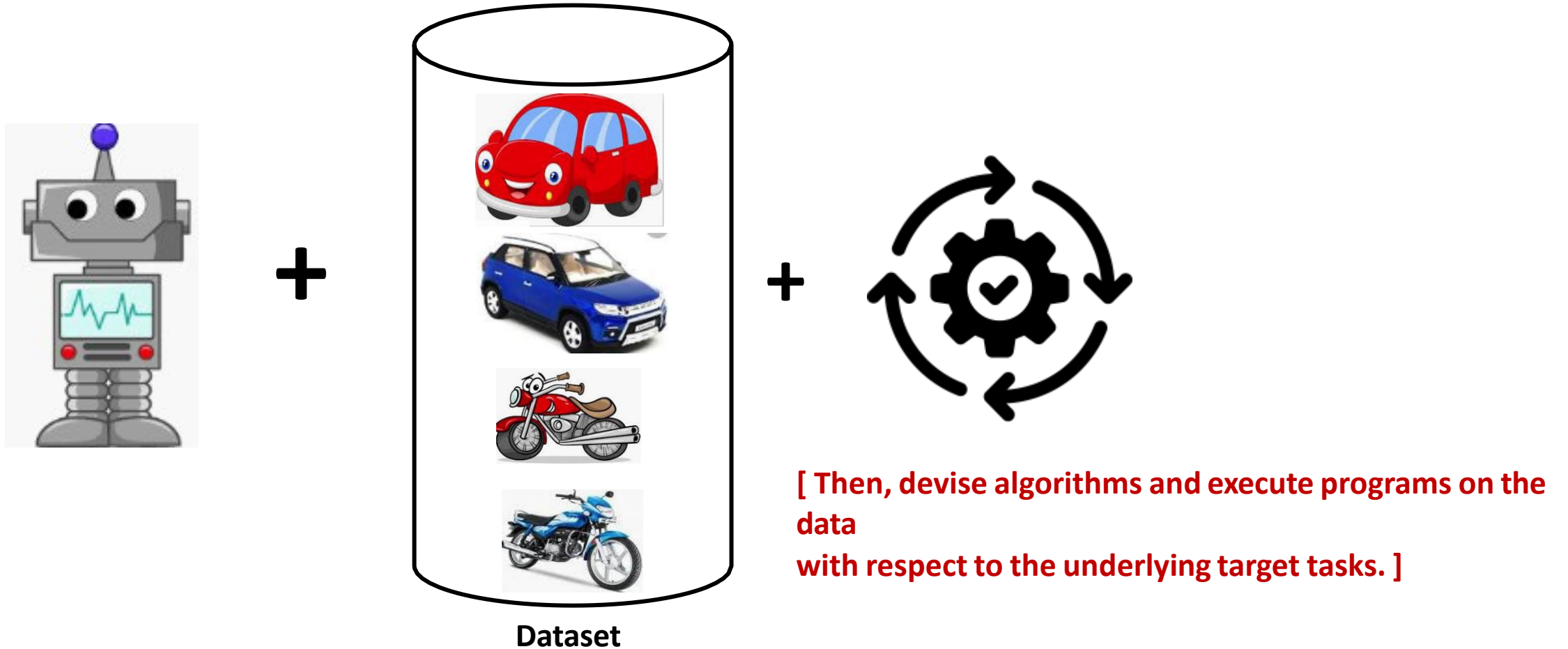


[
This is what we call a Data
or Training dataset.
So, we first need to provide a
training dataset to the machine.
]





What is Machine Learning?

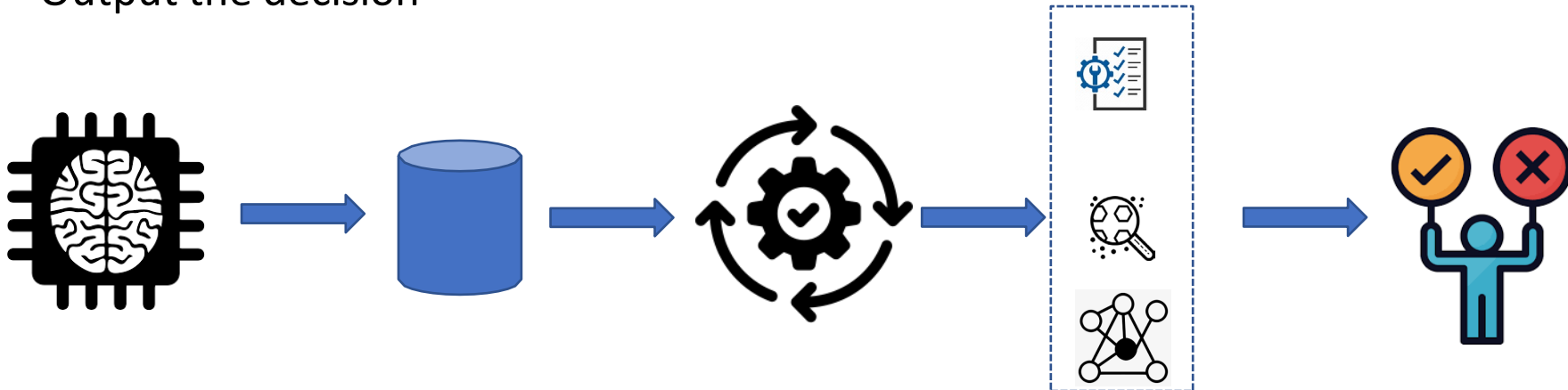




What is Machine Learning?

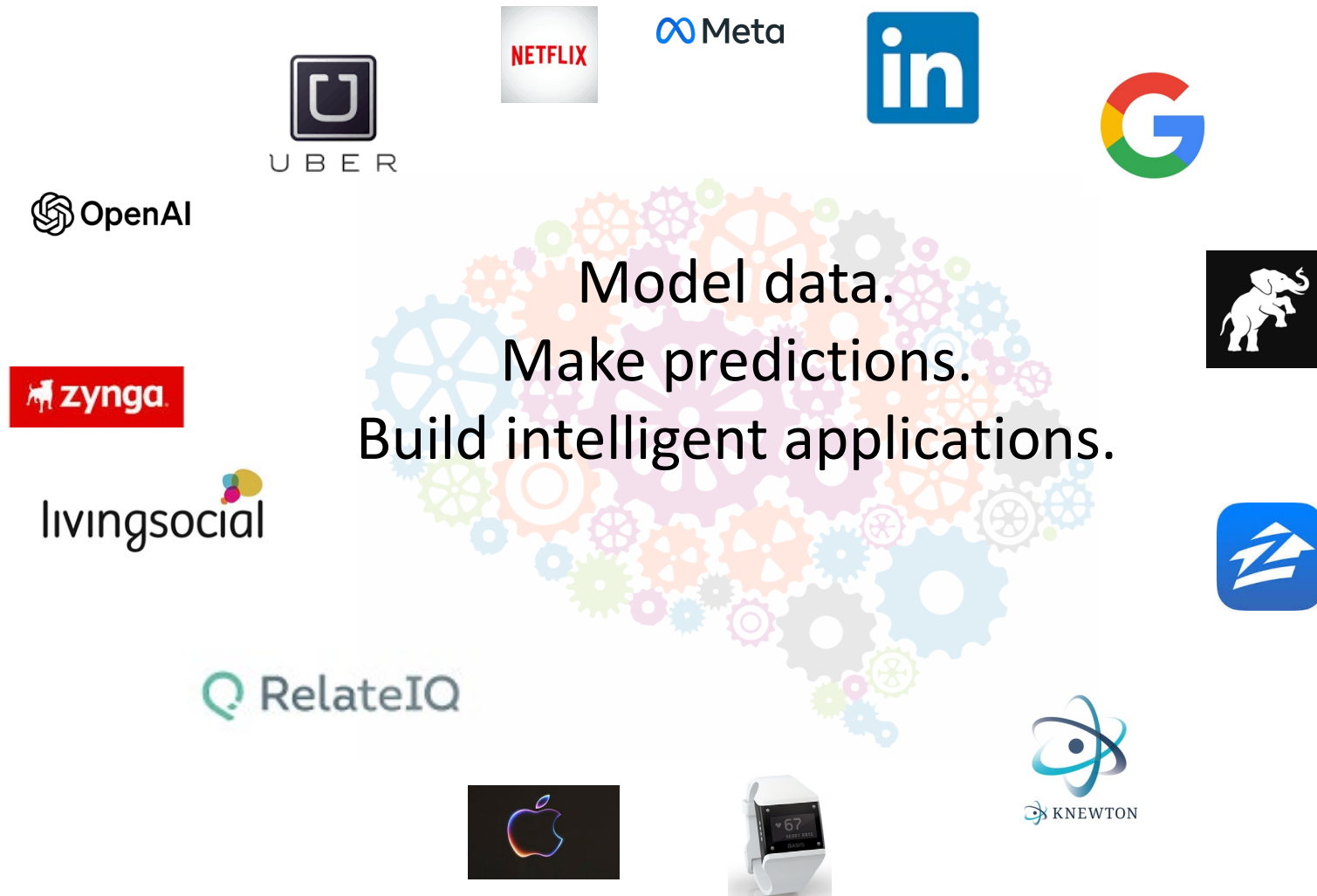
Given a machine learning problem

- Identify and create the appropriate dataset
- Perform computation to learn
 - Required rules, pattern and relations
- Output the decision





Why machine learning?





Defining the Learning Task

$\langle P, T, E \rangle$: Improve on task T , with respect to performance metric P , based on experience E (Tom Mitchell)

T : Playing checkers

P : Percentage of games won against an arbitrary opponent

E : Playing practice games against itself

T : Recognizing hand-written words

P : Percentage of words correctly classified

E : Database of human-labeled images of handwritten words

T : Driving on four-lane highways using vision sensors

P : Average distance travelled before a human-judged error

E : A sequence of images and steering commands recorded while observing a human driver.

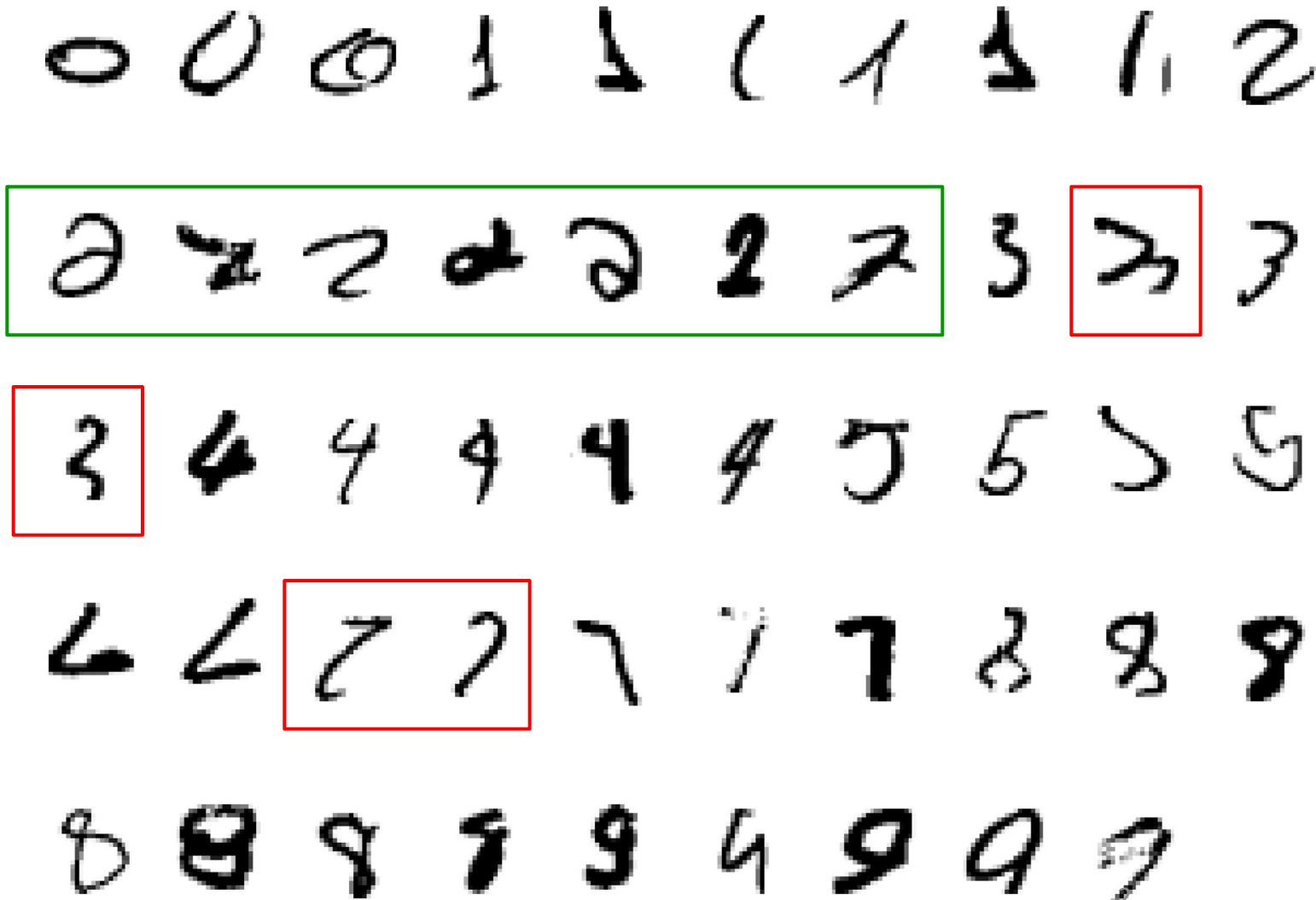
T : Categorize email messages as spam or legitimate.

P : Percentage of email messages correctly classified.

E : Database of emails, some with human-given labels

A CLASSIC EXAMPLE OF A TASK THAT REQUIRES MACHINE LEARNING:

Identifying the numeral 2 in a collection.

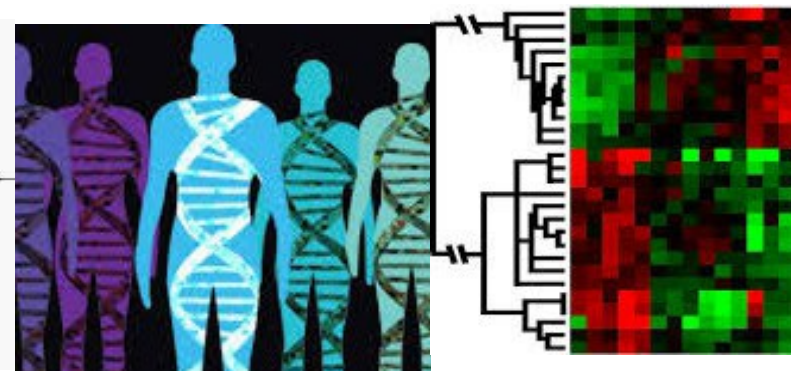
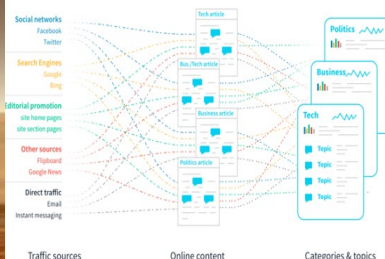




When Do We Use Machine Learning?

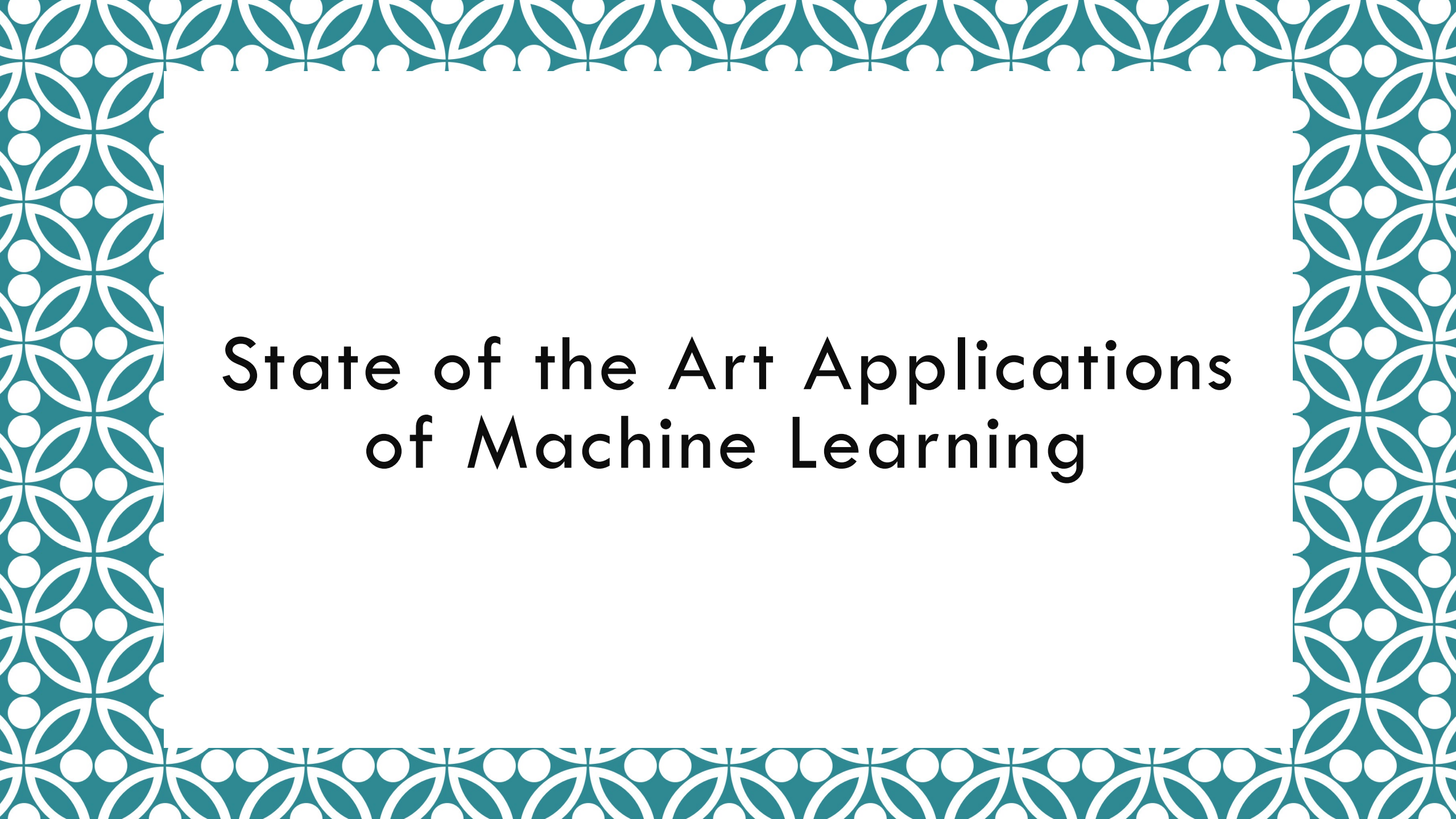
ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (NLP/image/speech recognition)
- Models must be customised (personalised medicine or Netflix)
- Models are based on huge amounts of data (genomics)



Learning isn't always useful:

- There is no need to “learn” to calculate payroll or “tax calculation” or “password authentication”

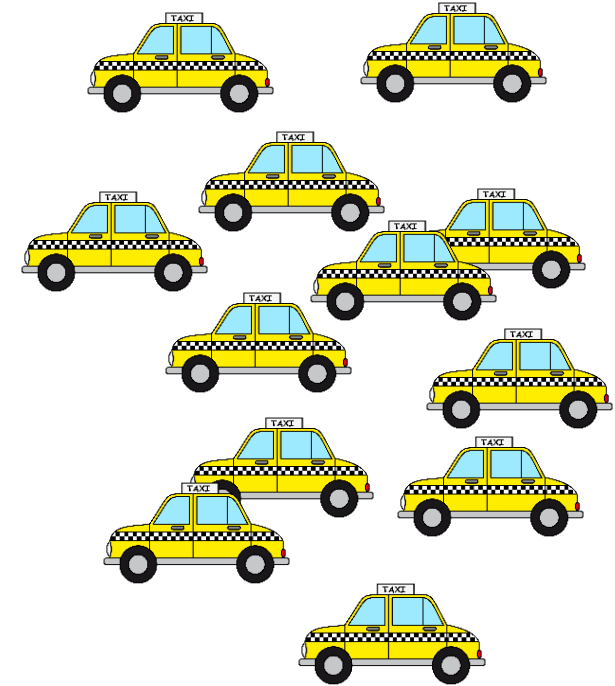
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State of the Art Applications of Machine Learning



Demand Prediction

■ How many taxis do I need in Brisbane on Friday evening?



Model



- Trx 1
- Trx 2
- Trx 3
- Trx 4
- Trx 5
- Trx 6
- ...



Model



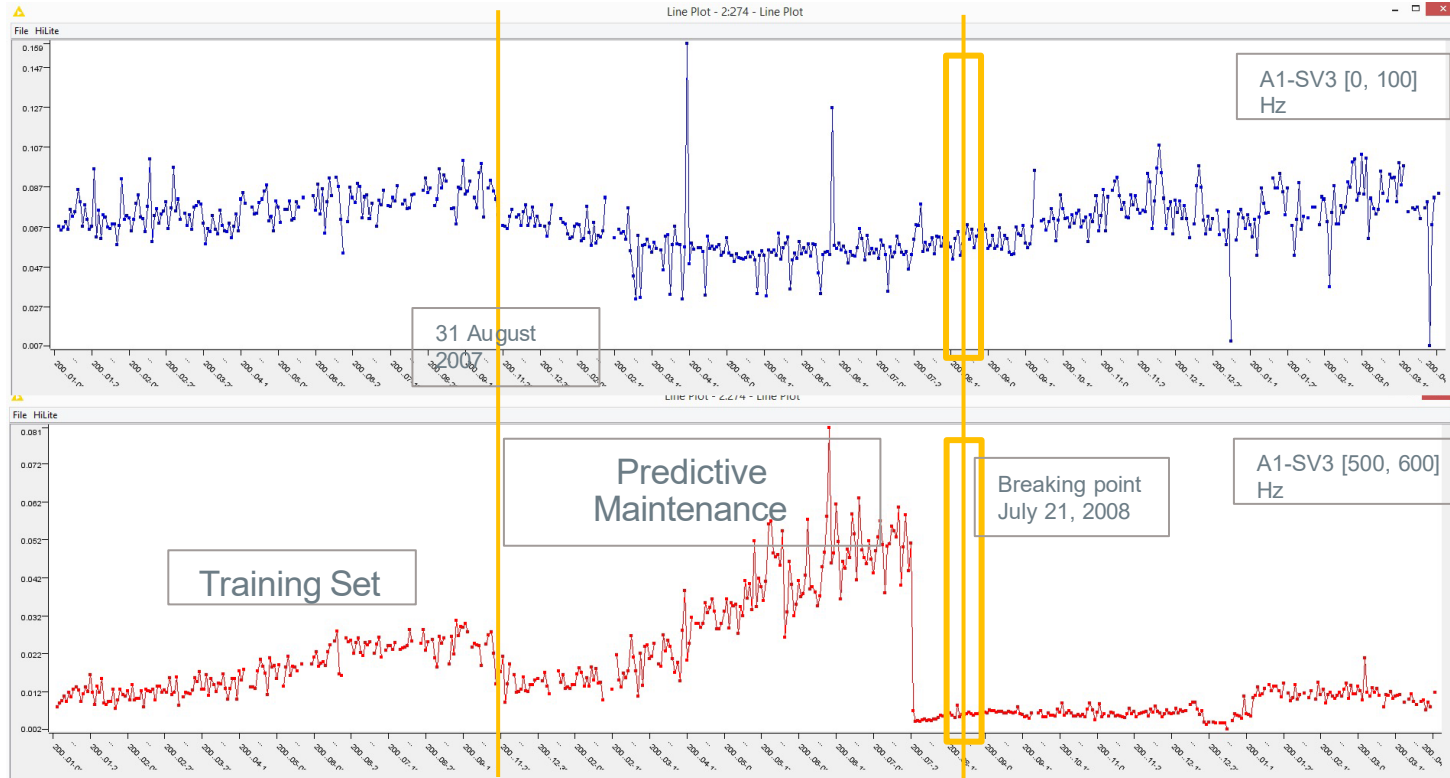
Suspicious Transaction

[illegible]



Anomaly Detection

Predicting mechanical failure as late as possible but before it happens



Dialog - 5270269:284:271 - Send Email

File | Attachments | Mail Host (SMTP) | Flow Variables | Memory Policy

To: rosana.silpo@kime.com

CC:

BCC:

Subject: 1st Level Alarm detected!

prediction
Mean(Sensor)[000, 700] + Amp A-3-5A1
RowID
currentIteration
currentColumnName
kime.workspace

Second Level Alarm has been detected today in Sensor and Frequency Band 00(SourcentColumnValue)00. Your Motor Administrator

Priority: Normal | Text | HTML

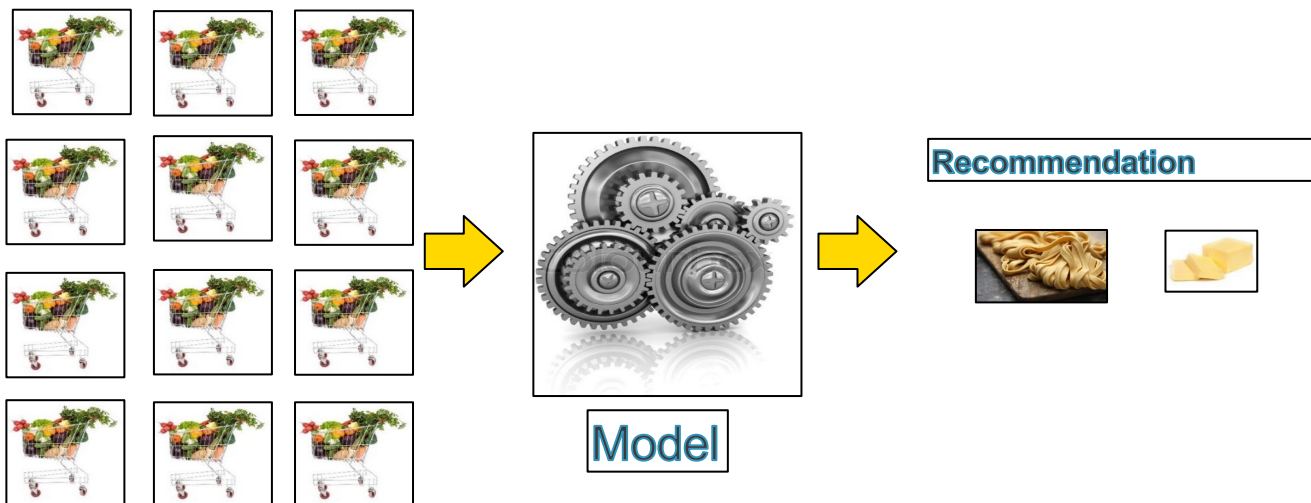
OK Apply Cancel ?



Only some Spectral Time Series shows the break down



Recommendation Engines/Market Basket Analysis: Amazon, Netflix



Inspired by your shopping trends



Related to items you've viewed [See more](#)



Natural Language Processing: Machine translation, sentiment analysis, topic modelling, text summarisation, Q&A, spam filtering (classification).

Real world example: The New York Times LDA analysis of 1.8M New York Times articles:



29 / 55



Samsung Galaxy S7 Edge G935A 32GB Unlocked - Gold Platinum

★★★★☆ 125 customer reviews | 606 answered questions

★★★★★ Beautiful phone from a wonderful seller!

By [Name] on May 29, 2017

Color: Gold | **Verified Purchase**

This practically new beautiful phone well exceeded my expectations!

★☆☆☆☆ One Star

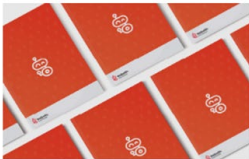
By [Name] on August 3, 2016

Color: Black Onyx | **Verified Purchase**

Very bad experience

Data driven marketing strategy technologies aligned to business goals

Robotic Marketer marketing strategies are smarter, more intuitive, and big data-based, with in-depth analysis of competitors and the market. Artificial intelligence (AI) provides more actionable insights into how to optimize your marketing plan. Robotic Marketer technology guarantees that you will receive the best robot-possible **marketing strategy** with the added benefits of a **marketing consultant**.



Big Data

Robotic Marketer is the future of marketing consulting. It digs beyond the algorithms of Google, for a more in-depth analysis and results-based **marketing strategy**. With so much data on the internet on your customers, competitors, influencers and industry, it is imperative that the robot digs deep and finds relevant information on what works and what does not for your company's marketing plan.

Artificial Intelligence

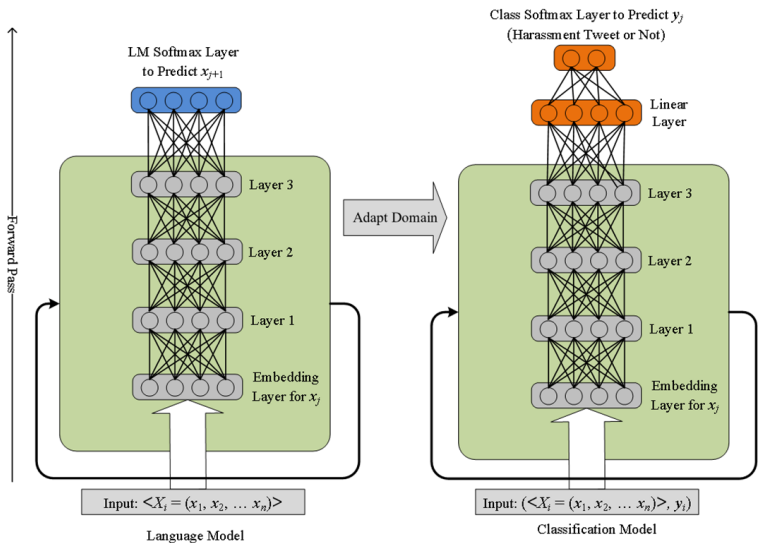
AI is the basis on which the robot is able to communicate, gather information, turn data into a real marketing plan, decipher where a marketing budget is best spent, search keywords and their relevancy to your **marketing strategy**, and provide the final, powerful, results-driven marketing strategy to you in minutes. AI will also be a big part of the marketing consulting process, starting with the marketing workshop.

Machine Learning

From adapting various **marketing strategies** and workshops, this robot has the knowledge to build your marketing plan. Machine learning is imperative to marketing consulting, allowing the robot to gain enough knowledge to build upon and use other applications like AI, big data and scraping to formulate the ultimate, powerful marketing strategy.

- Thousands of marketing strategies
- 100+ voice recorded marketing workshops
- 100's more as part of this process
- Founder has 25 years experience in marketing

Misogynistic Tweet Detection



Computer vision: Object detection, semantic segmentation, pose estimation, and almost every other task is done with ML.



Figure 4. More results of Mask R-CNN on COCO test images, using ResNet-101-FPN and running at 5 fps, with 35.7 mask AP (Table 1).



Instance segmentation - [▶ Link](https://www.youtube.com/watch?v=YGO2lwA9rig)



DAQUAR 1553

What is there in front of the sofa?

Ground truth: table

IMG+BOW: table (0.74)

2-VIS+BLSTM: table (0.88)

LSTM: chair (0.47)



COCOQA 5078

How many leftover donuts is the red bicycle holding?

Ground truth: three

IMG+BOW: two (0.51)

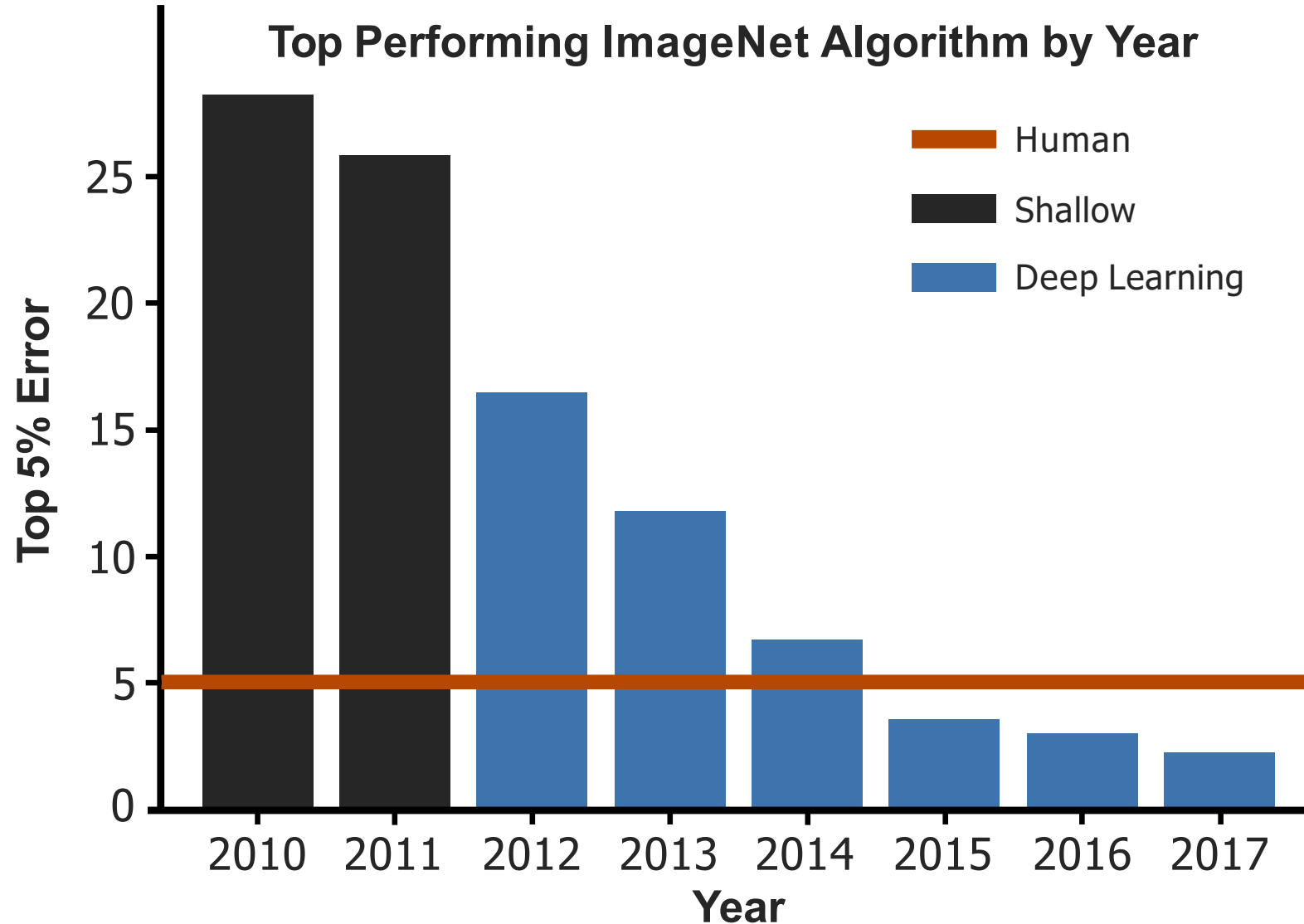
2-VIS+BLSTM: three (0.27)

BOW: one (0.29)



ImageNet Challenge

Top Performing ImageNet Algorithm by Year



Convolutional Neural Network

- First used in 2012 at scale
- Outperforms humans on the same task



mite

container ship

motor scooter

leopard

← true image label

mite
black widow
cockroach
tick
starfish

container ship
lifeboat
amphibian
fireboat
drilling platform

motor scooter
go-kart
moped
bumper car
golf cart

leopard
jaguar
cheetah
snow leopard
Egyptian cat

assigned labels by machine



grille

mushroom

cherry

Madagascar cat

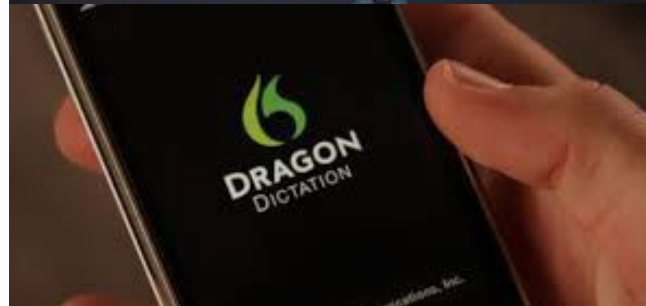
← true image label

convertible	agaric	dalmatian	squirrel monkey
grille	mushroom	grape	spider monkey
pickup	jelly fungus	elderberry	titi
beach wagon	gill fungus	Staffordshire bullterrier	indri
fire engine	dead-man's-fingers	currant	howler monkey

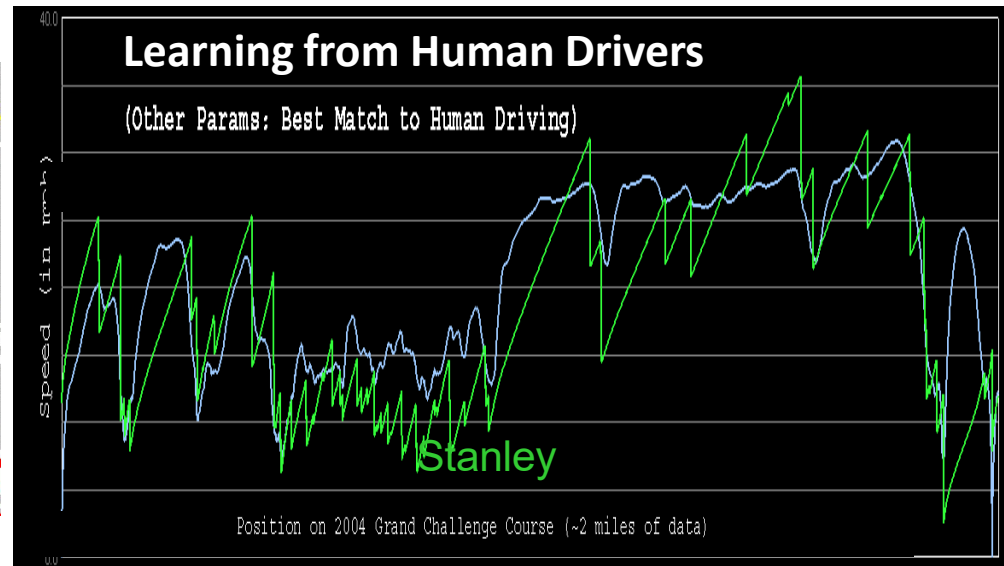
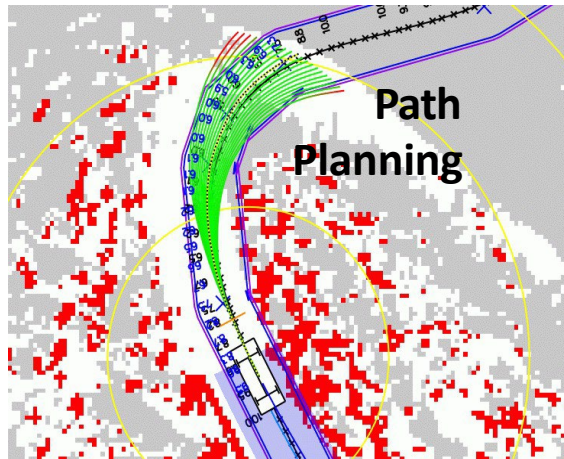
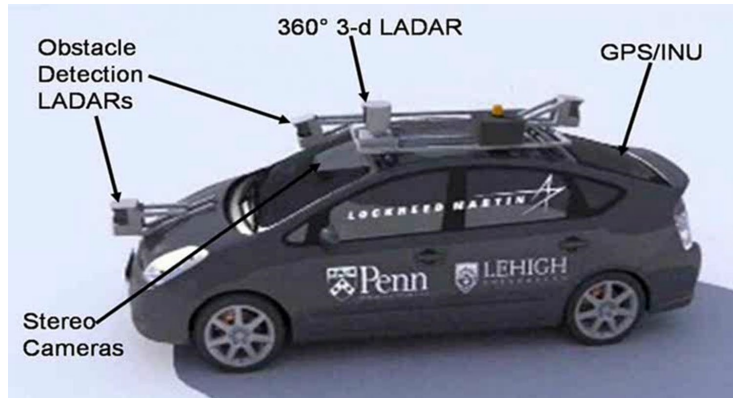
assigned labels by machine



Speech Technology: Speech-to-text, personal assistants, speaker identification...



Autonomous car technology





Generative AI

Generative AI used to plan Tesla Cybertruck explosion outside Trump Hotel: Las Vegas Police

AP |

Jan 08, 2025 03:25 AM IST



Read Now

TECH

OpenAI announces GPT-4, claims it can beat 90% of humans on the SAT

PUBLISHED TUE, MAR 14 2023-1:42 PM EDT | UPDATED TUE, MAR 14 2023-2:32 PM EDT



Kif Leswing
@KIFLESWING

SHARE



MOTHERBOARD
TECH BY VICE

The New GPT-4 AI Gets Top Marks in Law, Medical Exams, OpenAI Claims

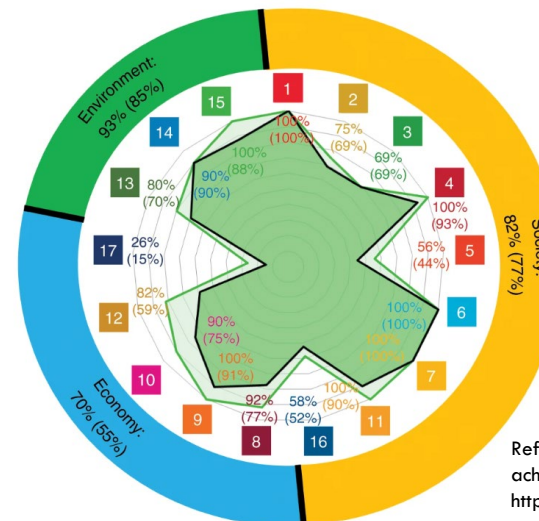


From Applications to Societal Impact: Sustainable Development Goals

- 17 global goals
- Examples of Machine Learning contributions
 - 3 Health – Disease detection, medical imaging
 - 4 Education – Personalisation
 - 6 Water – Quality monitoring (anomaly detection)
 - 7 Energy – Demand forecasting
 - 11 Cities – Traffic prediction
- The success of Machine Learning should be measured not only by accuracy, but by its societal impact.

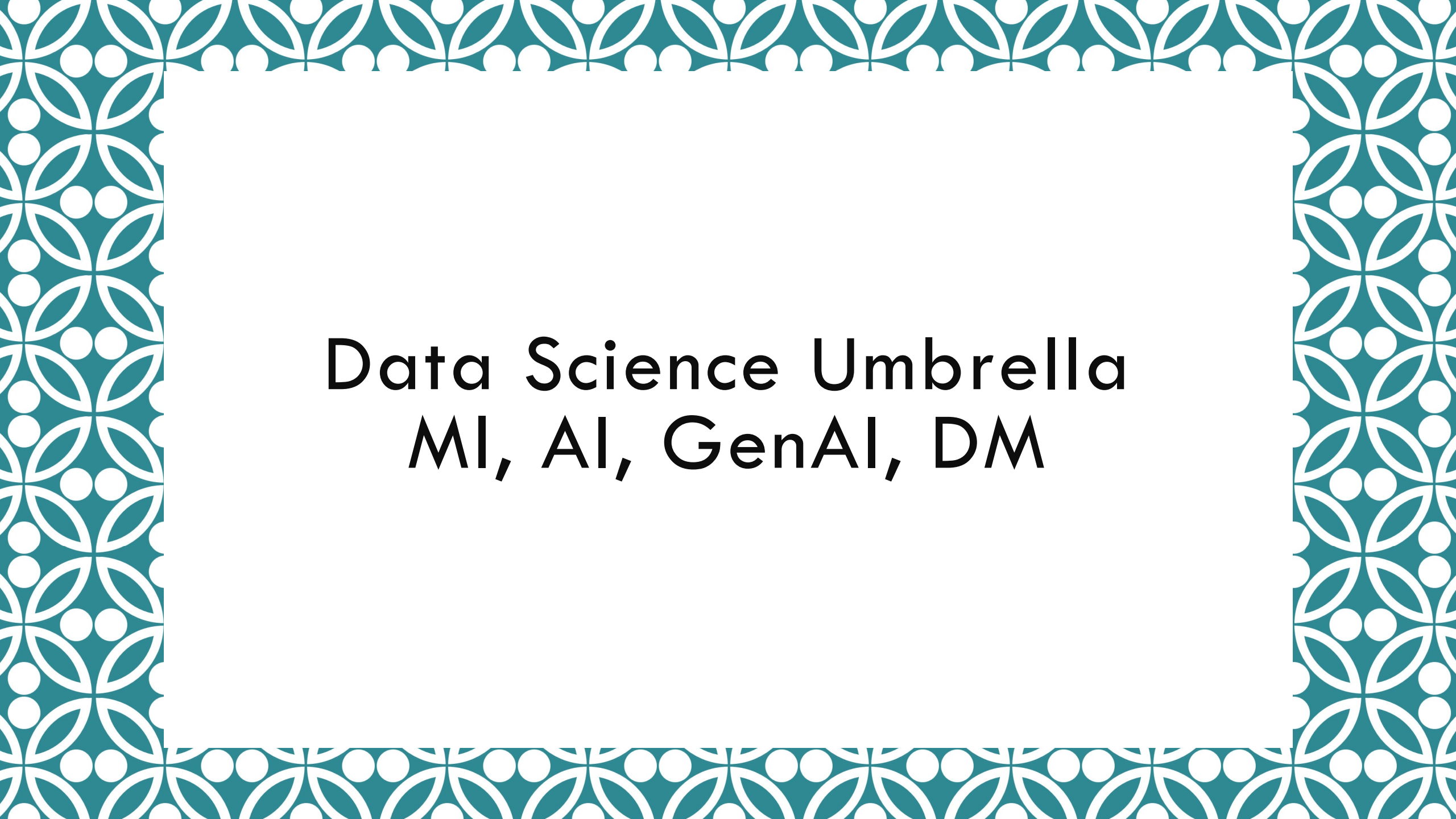


Source: <https://sdgs.un.org/goals>



AI/ML as an enabler

Ref: Vinuesa, R., Azizpour, H., Leite, I. *et al.* The role of artificial intelligence in achieving the Sustainable Development Goals. *Nat Commun* **11**, 233 (2020). <https://doi.org/10.1038/s41467-019-14108-y>

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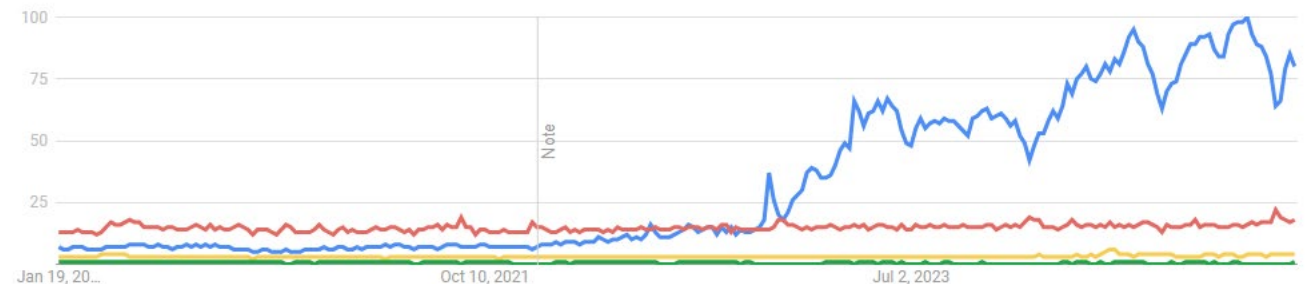
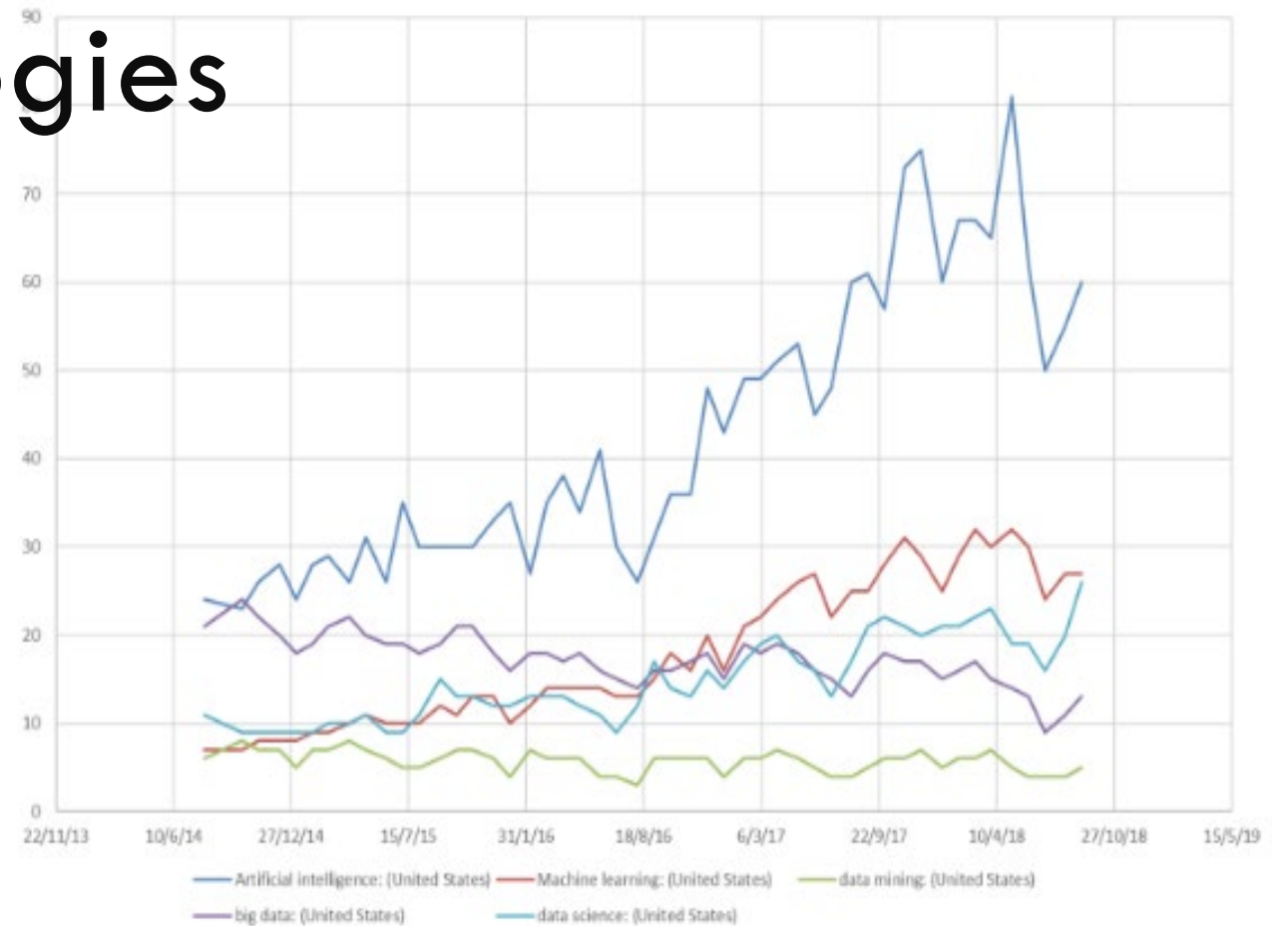
Data Science Umbrella

MI, AI, GenAI, DM



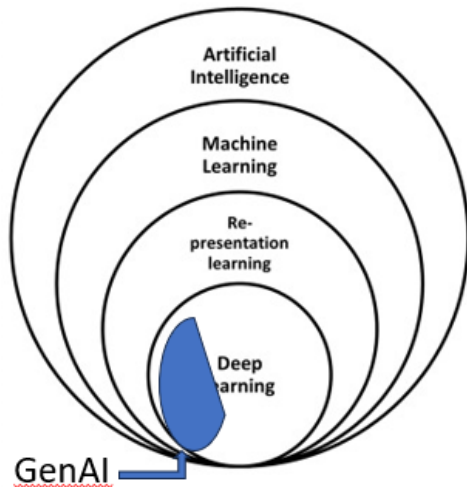
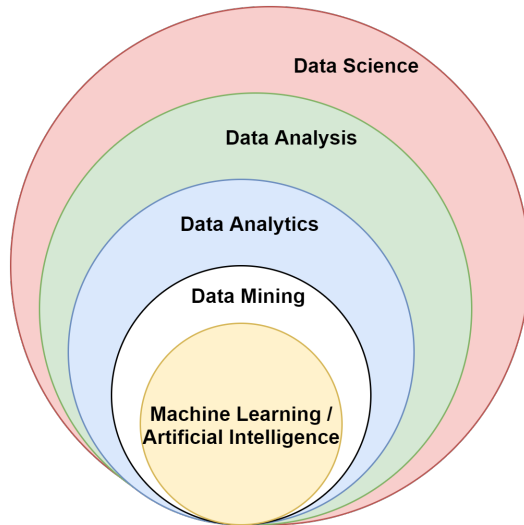
Related terminologies

- Data Analysis
- Data Analytics
- Data Mining (DM)
- Big Data
- Machine Learning (ML)
- Artificial Intelligence (AI)
- Generative AI (GenAI)
- Data Science (DS)





Data Science Umbrella



- Data science acts as an umbrella for all these.
- Data Analysis: **Process** of analysing the data that leads to some insights. The process also includes the examination (exploring), transformation (converting), and arrangement (organising) of data. The insights help some **decision-making**.
- Data analytics: **Tools and techniques** used to perform data analysis. This also includes the data collection, storage, and organization.
- Data mining: **Process** of the systematic extraction of meaningful information from the data.
- ML and AI are used in Data Science, but they are generic techniques applied in many other domains.
 - Robotics
 - Signal processing
 - Psychological research
 - Chemometrics, etc.,

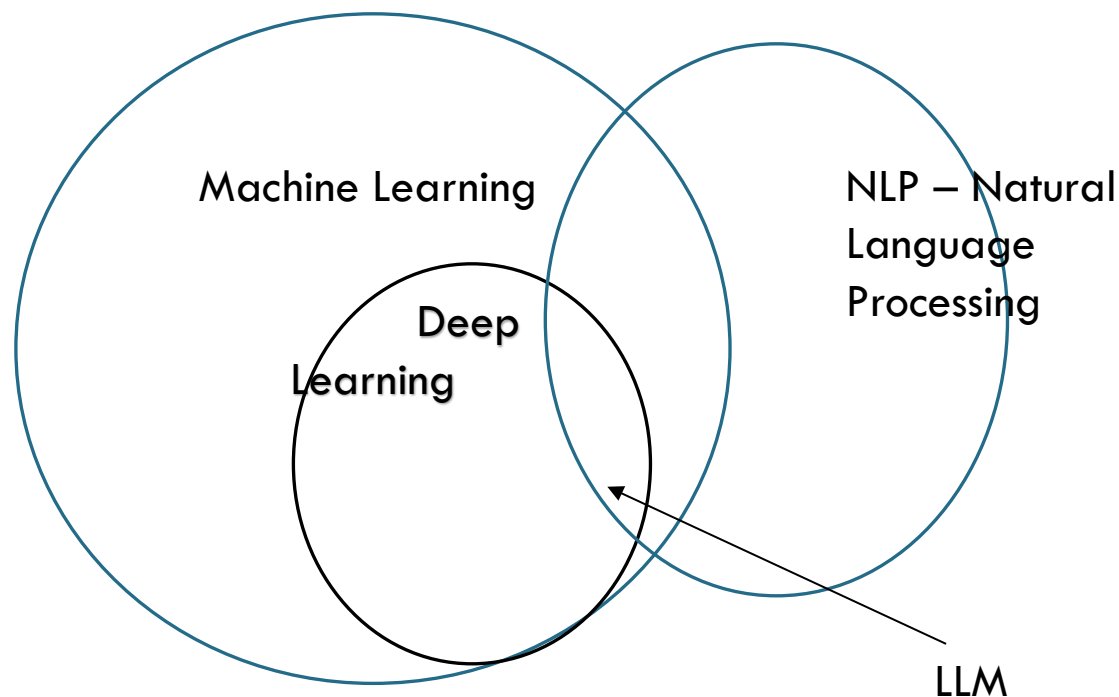


Machine Learning / Artificial Intelligence

- **Machine learning** is the ability of machines to learn from the data through algorithms.
- These algorithms help to get insights out of the data.
- Different machine learning algorithms exist to deal with different learning requirements.
- **Artificial Intelligence** is the ability of machines to simulate the human brain and the attempt to make machines intelligent.
- **Deep learning** is the type of machine learning method based on neural networks with large and deep architectures that attempt to mimic human cognitive and reasoning capabilities.
- **Generative AI (GenAI)** is a branch of artificial intelligence that creates new content, such as text, images, music, or code, by learning patterns and structures from existing data.



Large Language Models (LLMs) (e.g. GPT)



Language models are AI models to mimic human NLP capability.

Language models based on deep learning are

- trained to learn a probability distribution over sequences of words,
- to predict the next word or words in a text based on the preceding words and
- answer questions based on their stored knowledge.

One of the models behind ChatGPT (gpt-3.5-turbo) has 175 billion parameters.

GPT-4 has an unknown number of parameters.

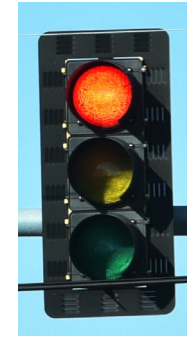


Many (but not all) AI systems are ML.

These are both AI. Which one is ML?



Option A



red pixels = 142
yellow pixels = 30
green pixels = 35

RED LIGHT
PREDICTED

Option B



GPUs vs. CPUs

CPU, like a motorcycle

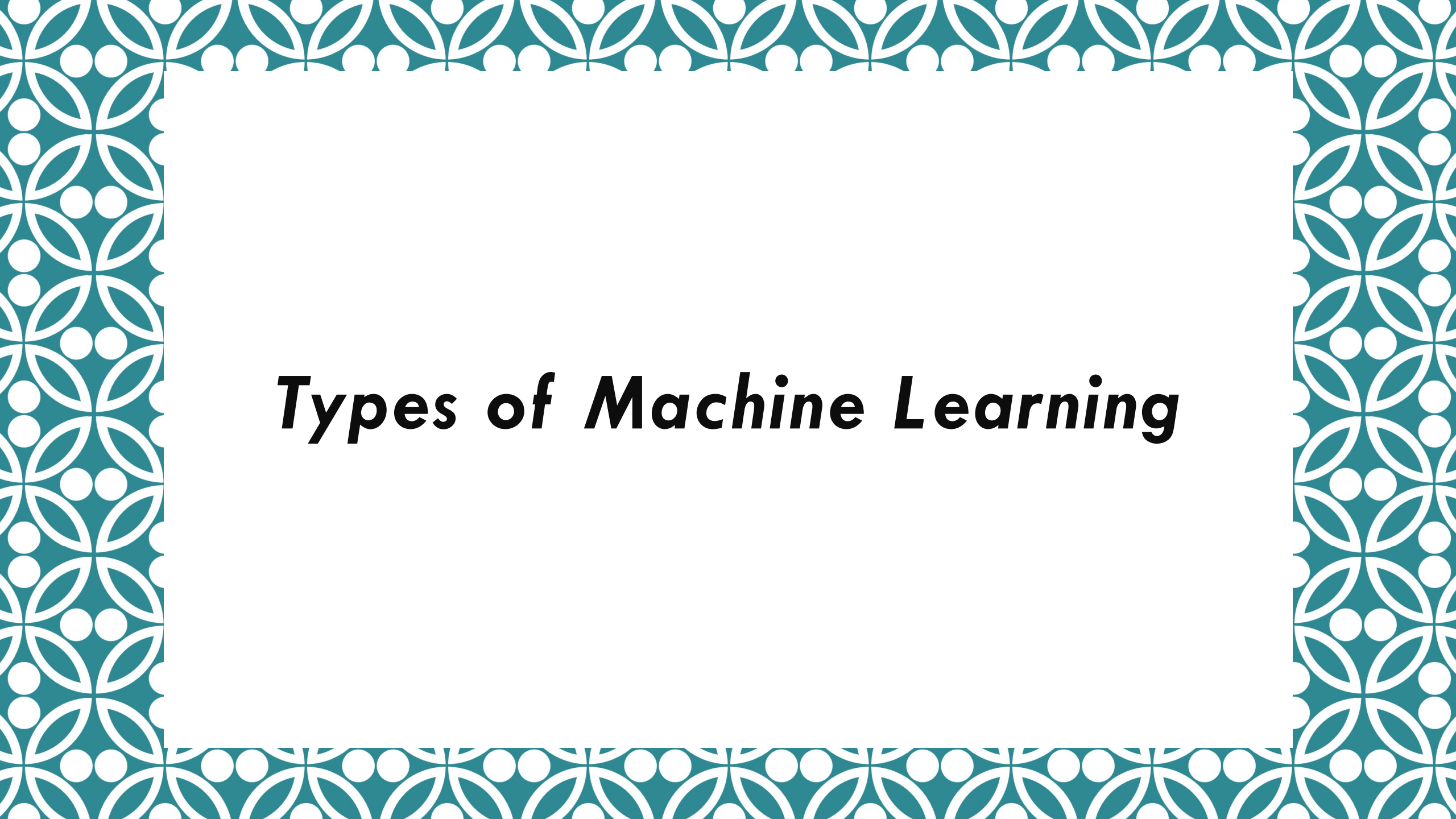


Quick to start, top speed
not shabby

GPU, like an airplane



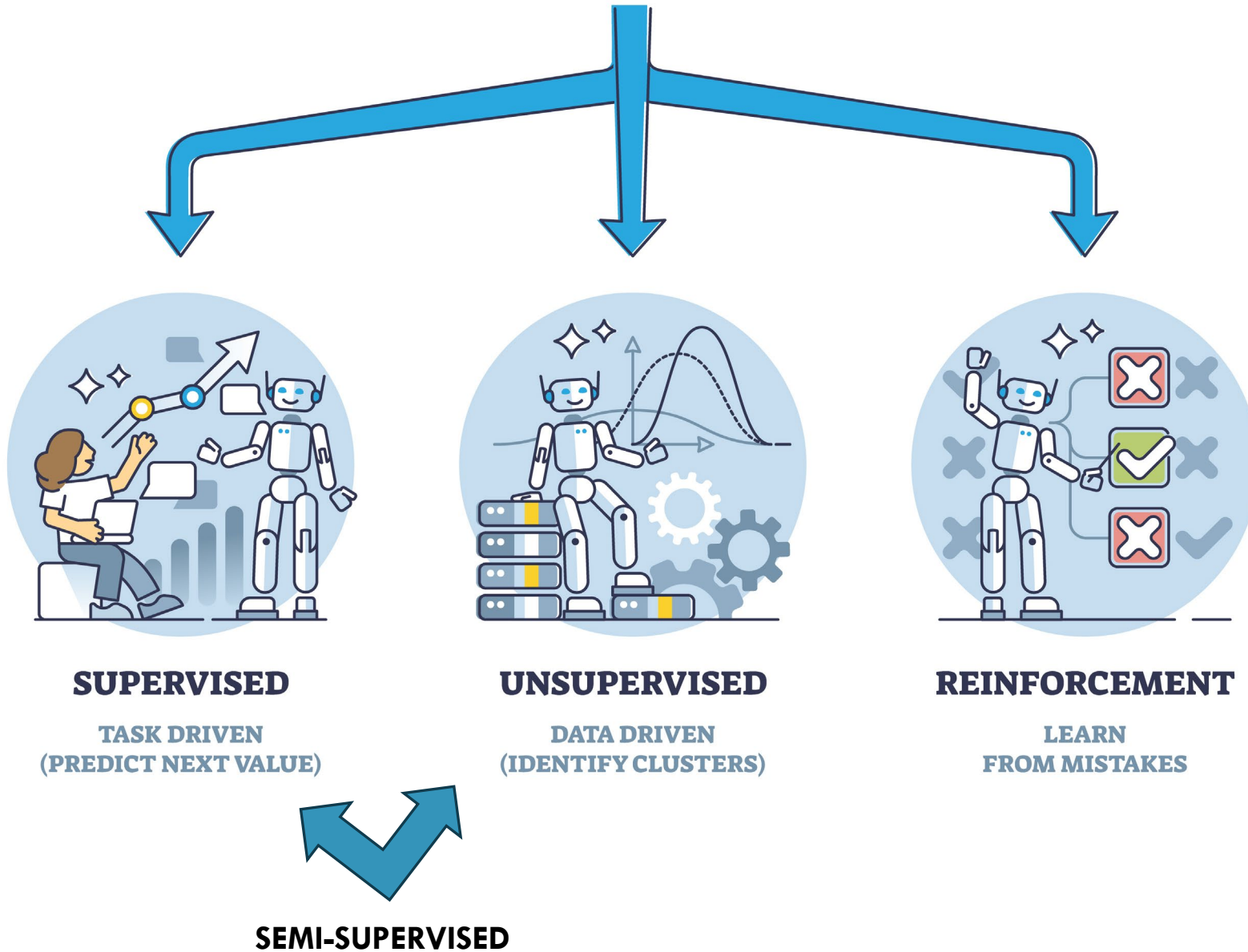
Takes forever to get off the
ground, but super-fast
once flying

The slide features a decorative border with a repeating geometric pattern of teal and white shapes, including circles and leaf-like motifs, framing the central content area.

Types of Machine Learning

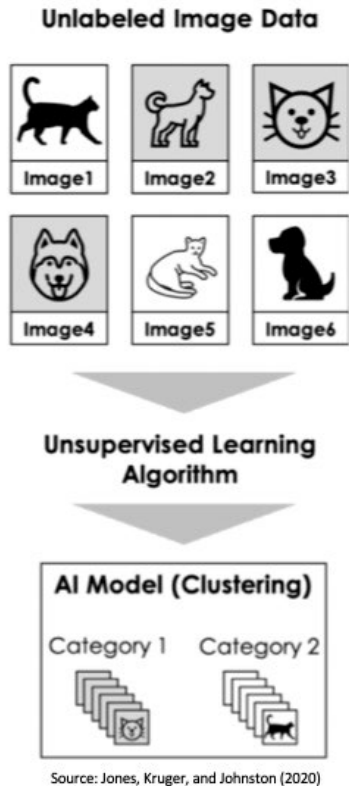


Types of ML.

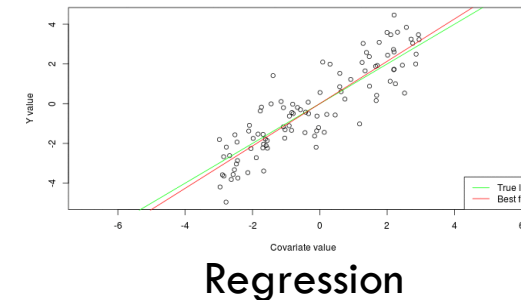
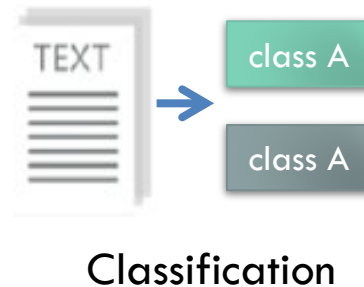
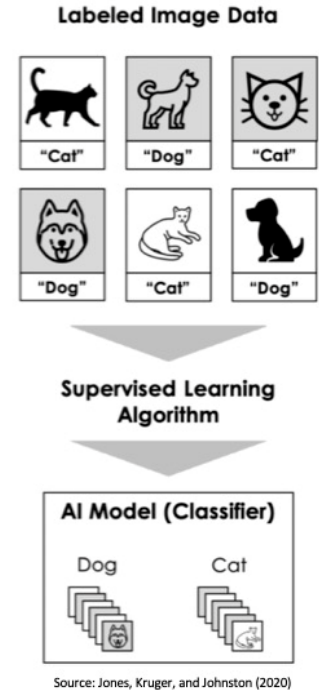




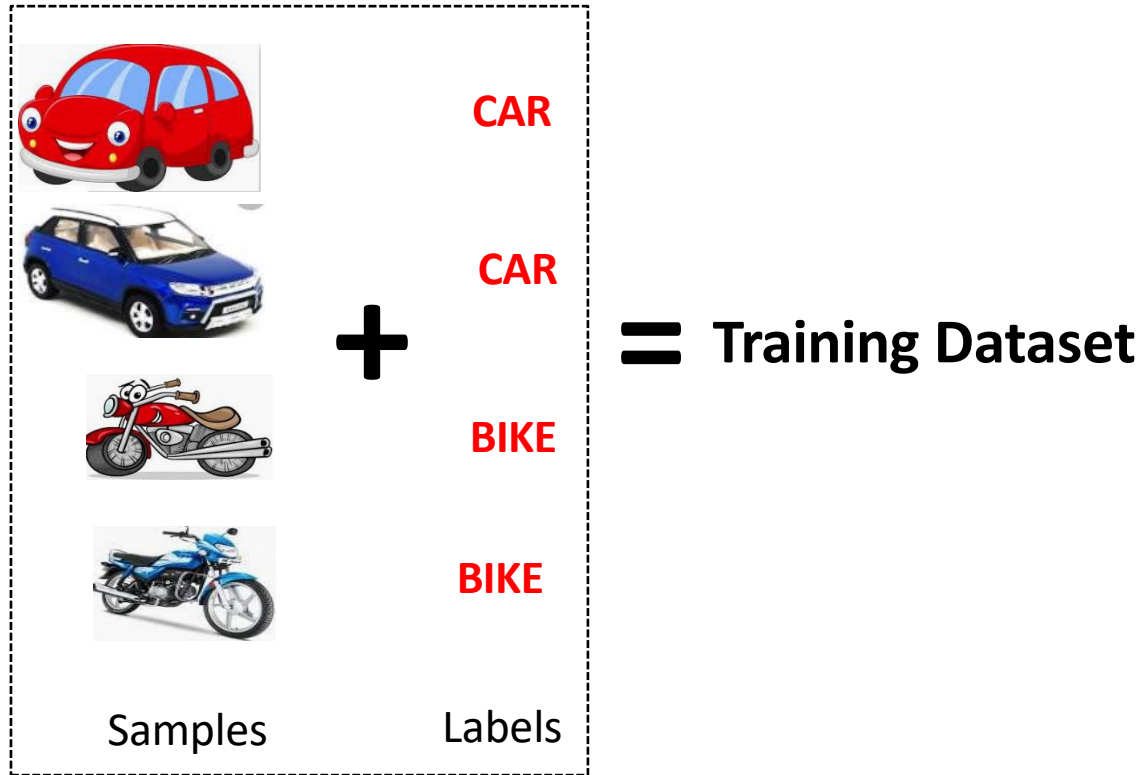
Supervised vs. Unsupervised Learning



- Unsupervised learning (**clustering**)
 - The class labels of training data are unknown
 - Discover common **patterns** in **unlabeled** data
 - Example: **cluster** similar documents based on text
- Supervised learning (**predictive modelling**)
 - Supervision: The training data (observations, measurements, etc.) are accompanied by labels indicating the class of the observations
 - Learning with a **labeled training** set
 - Example: (1) email **classification** with already labelled emails (2) loan amount prediction (**regression**) using the historical data
- Reinforcement learning
 - Learn to act based on feedback/reward: win or lose
 - Example: learn to play Go



What is Supervised Learning?



$$f(\text{Database}, \text{Sample}) =$$

$$f(\text{Database}, \text{Yellow Sports Car}) = \text{CAR}$$

Classification

$$f(\text{Database}, \text{Yellow Bus}) =$$

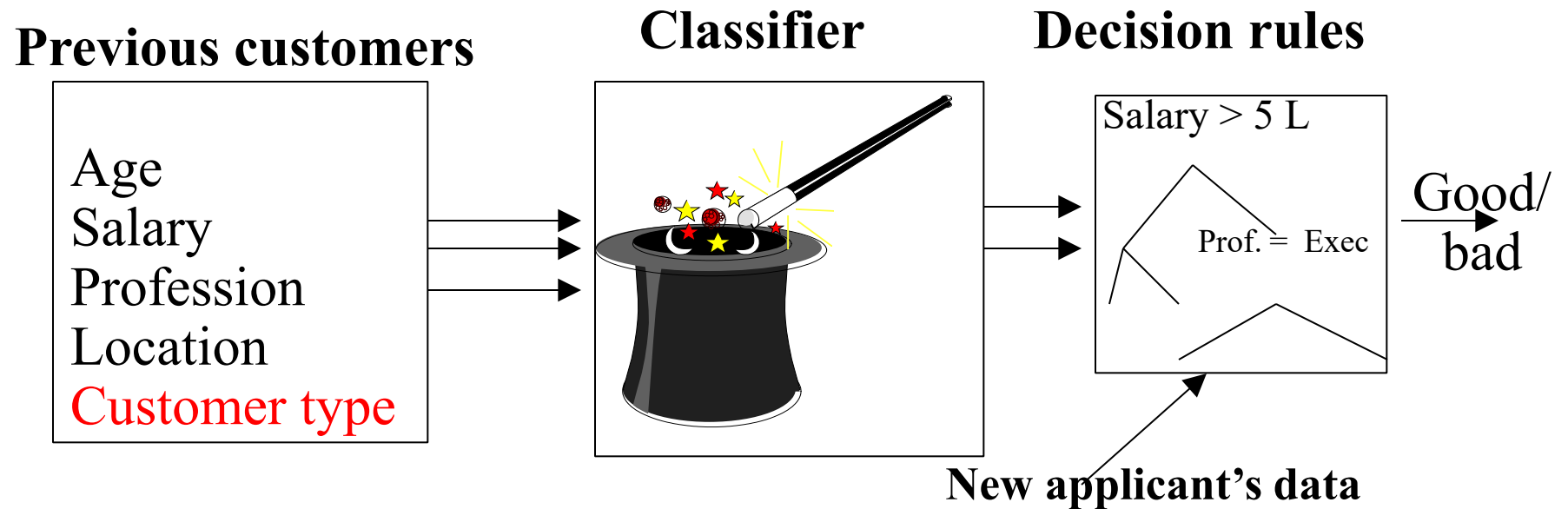
Predefined classes

- The model will produce output only from the labels defined in the dataset.
- For example, even if we input a bus, it will produce either CAR or BIKE.



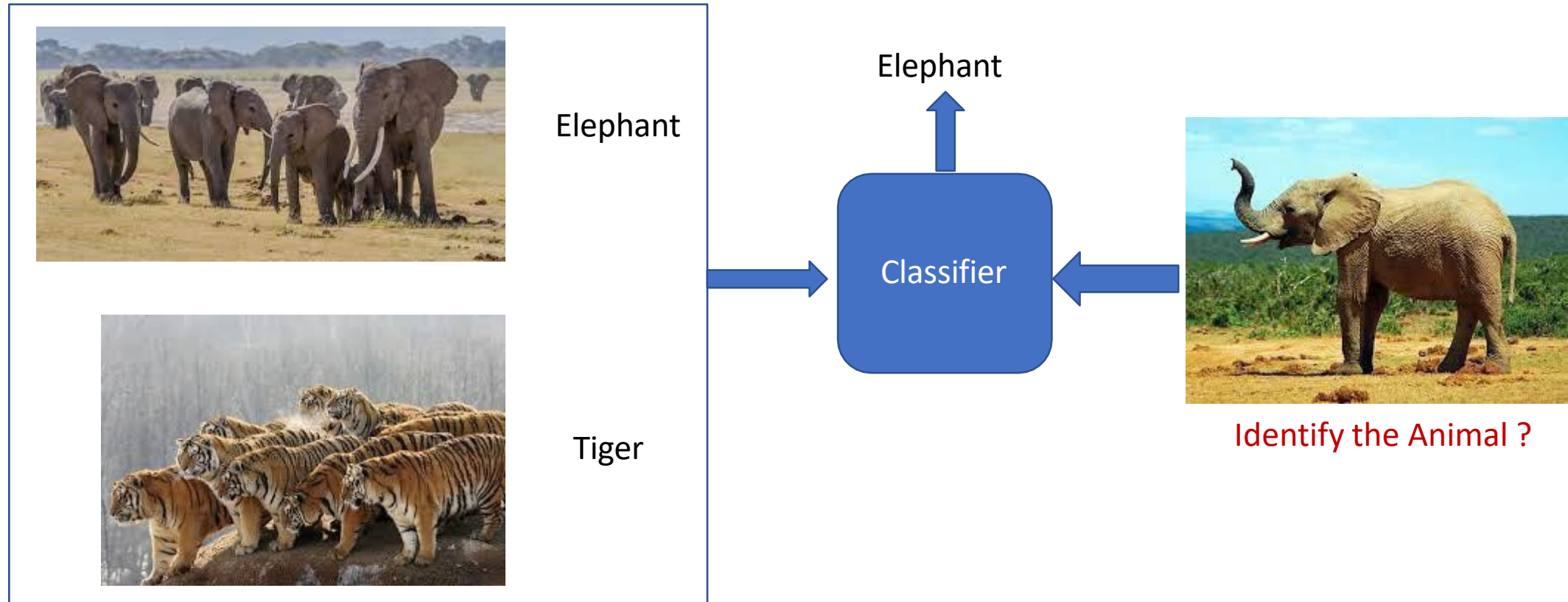
Supervised learning: Classification

- Given historical data about customers and payments, predict new applicant's loan eligibility.





Example: Classification



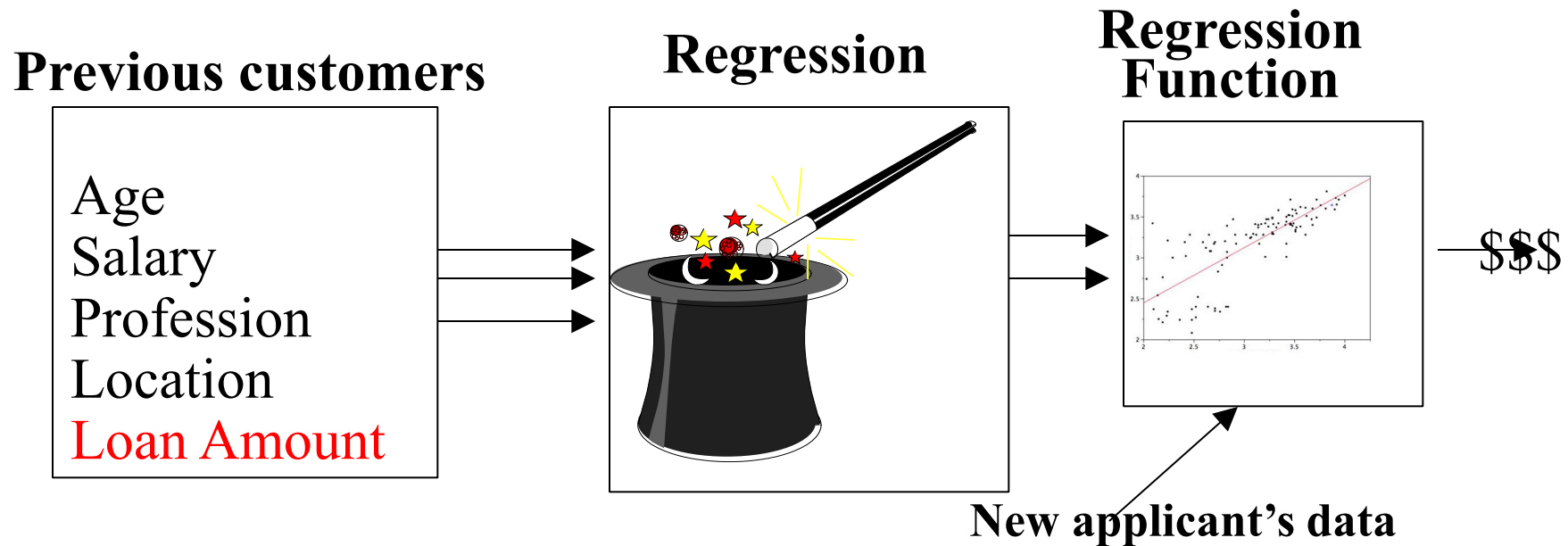
Dataset

- Predict the output values of the function as categorical values



Supervised Learning: Regression or Value Prediction

- Given historical data about customers and payments, predict new applicant's approved loan amount.





Example: Regression



Dataset

Regression

$$\bullet f(\text{house icon}) = 20500.50$$

- Predict the output values of the function as continuous real values



What is Unsupervised Learning



~~CAR~~



~~CAR~~



~~BIKE~~



~~BIKE~~



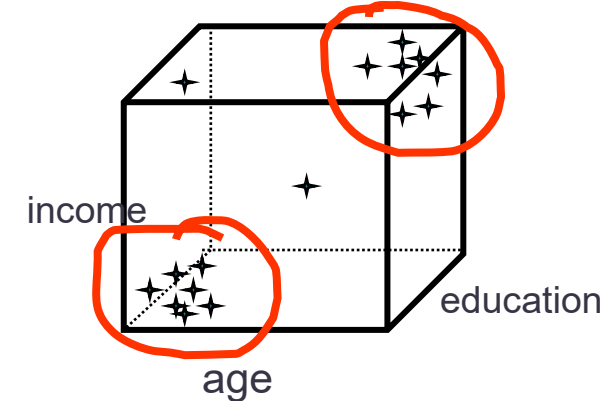
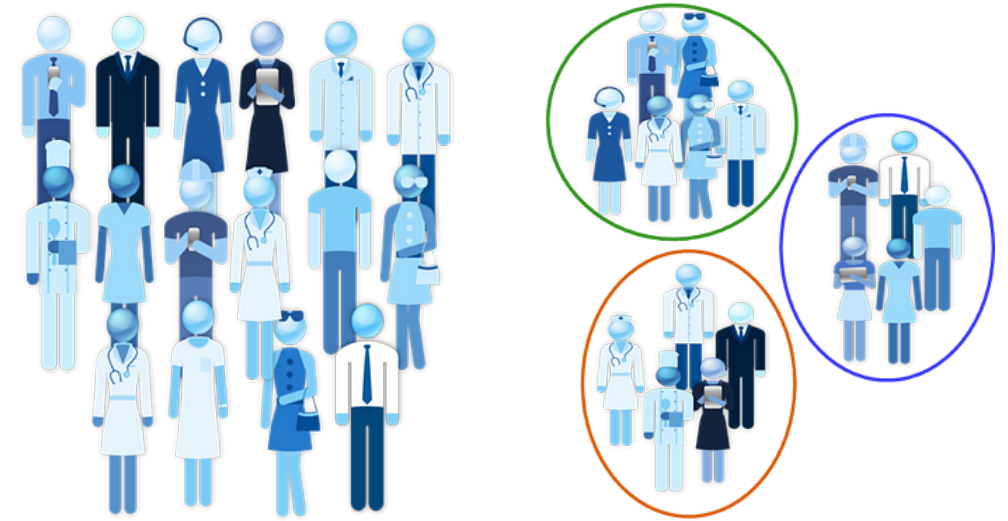
Clustering

Dataset

The task is to identify the patterns based on similarity and find subgroups or clusters such that

Data points in one cluster are more similar to one another.

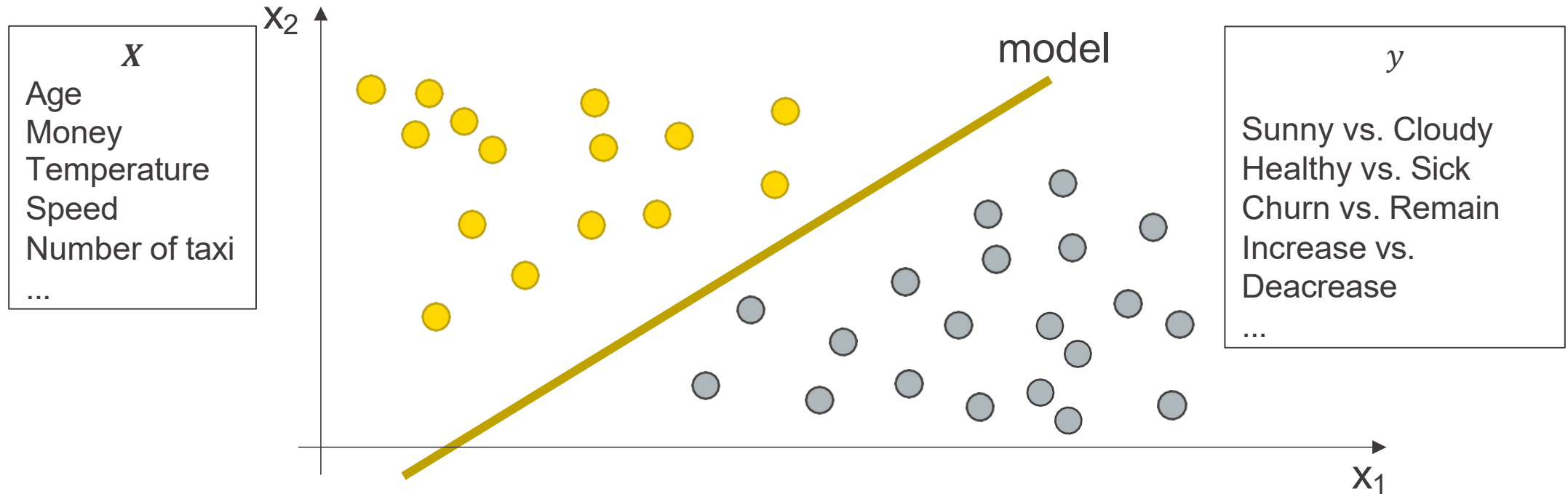
Data points in separate clusters are less similar to one another.





Supervised Learning

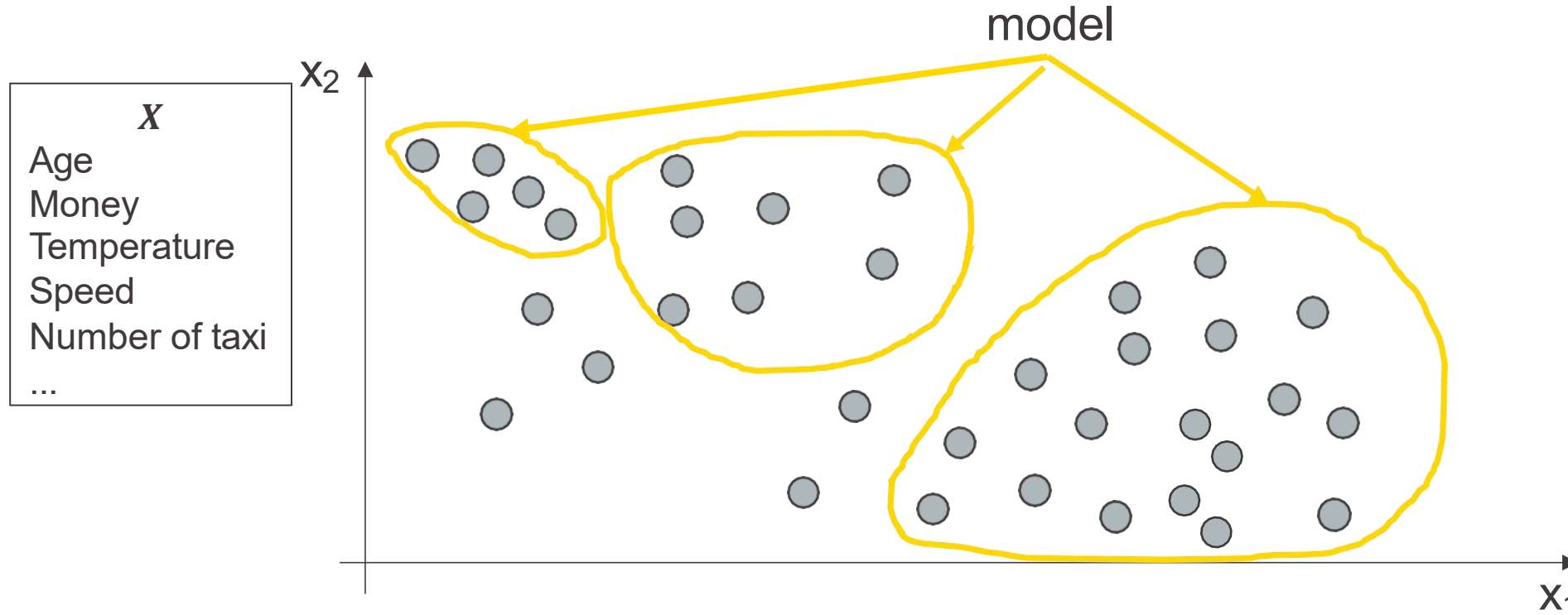
- $\mathbf{X} = (x_1, x_2)$ and $y = \{yellow, gray\}$
- A training set with many examples of $(\mathbf{X}, y)$
- The model learns on the examples of the training set to produce the right value of y for an input vector $\mathbf{X}$





Unsupervised Learning

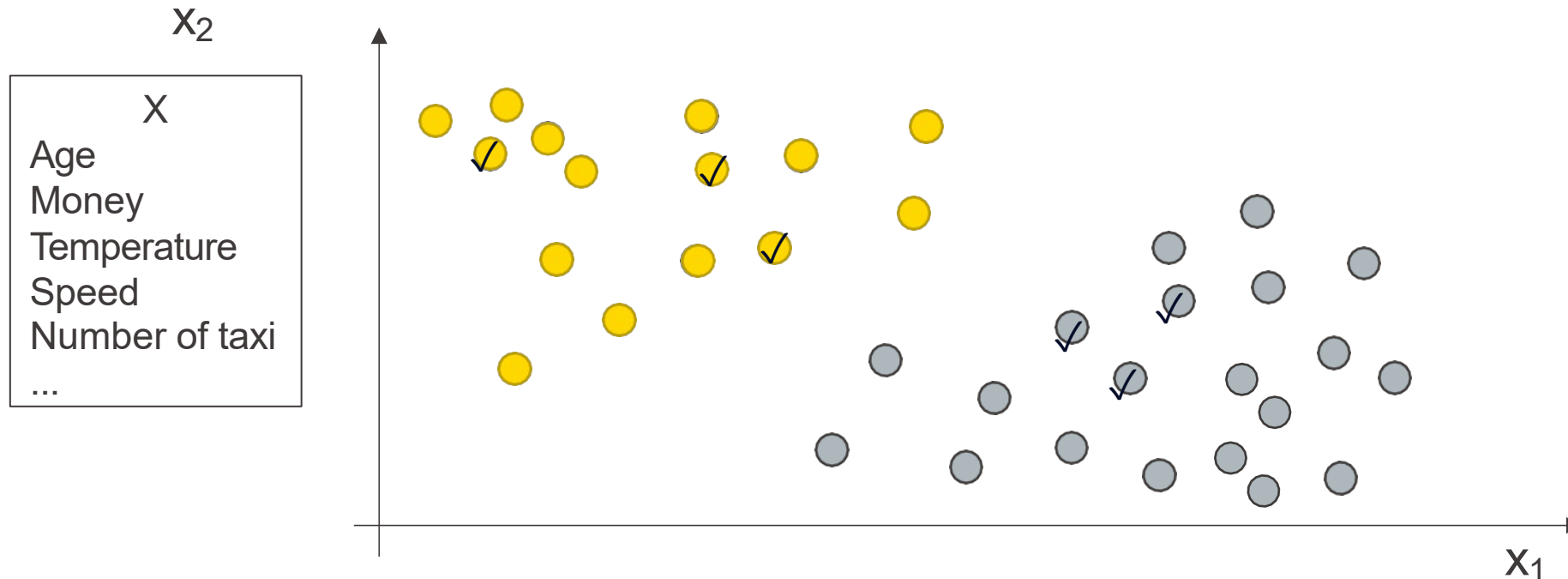
- $X = (x_1, x_2)$ and $y = \text{~~\{yellow, gray\}}~~}$
- A training set with many examples of $(X, \text{~~y~~})$
- The model learns to group the examples X of the training set based on similarity (closeness) or probability





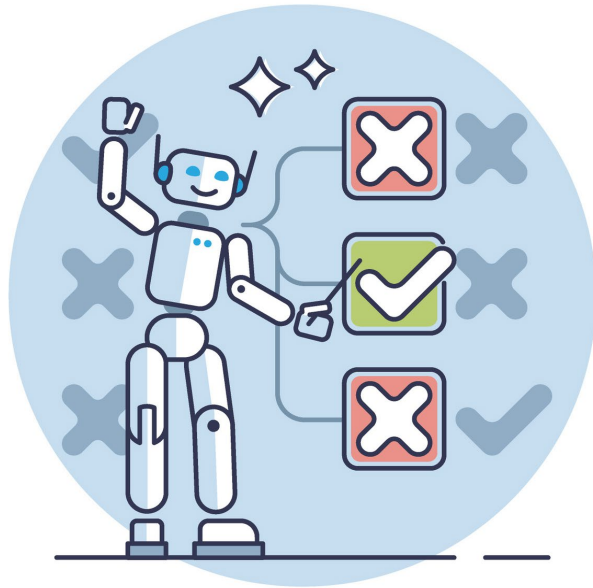
Semi-Supervised Learning

- $X = (x_1, x_2)$ and $y = \{yellow, gray\}$
- A training set with many examples of $(X, \text{~~y~~})$ and some samples (X, y)
- The model labels the data in the training set using a modified unsupervised learning procedure





Reinforcement Learning

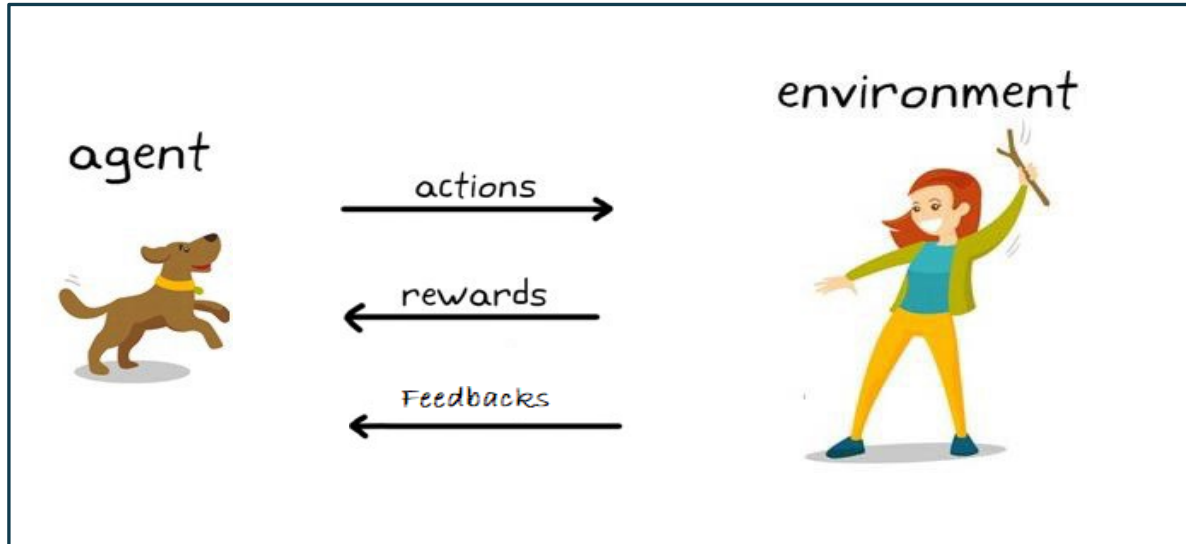


REINFORCEMENT

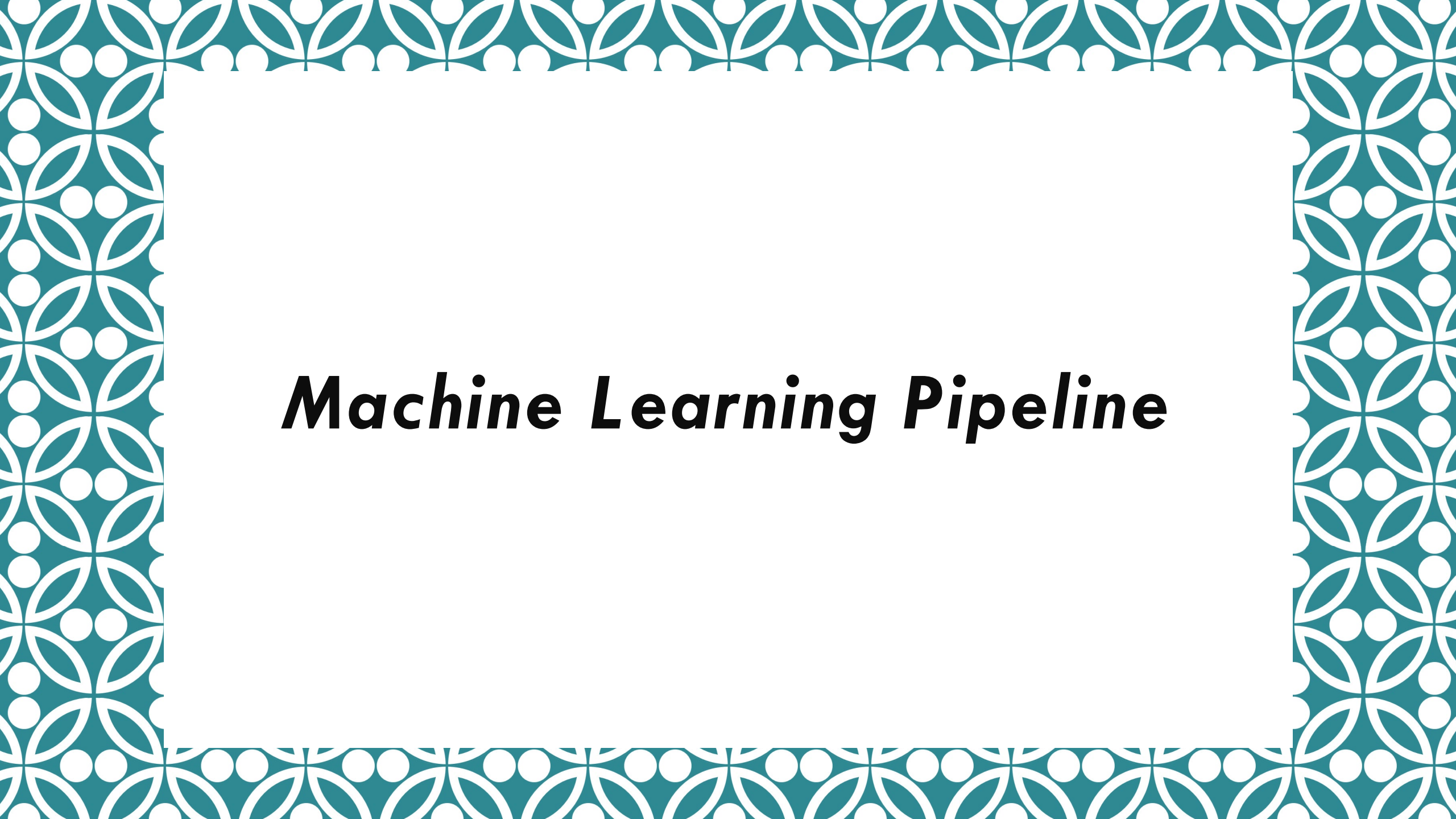
**LEARN
FROM MISTAKES**

- ML model is given an environment to interact with (data is gathered through interaction with environment)
- Model takes actions and then is given a "reward" if the actions led to success – trial and error learning

Reinforcement learning: Examples



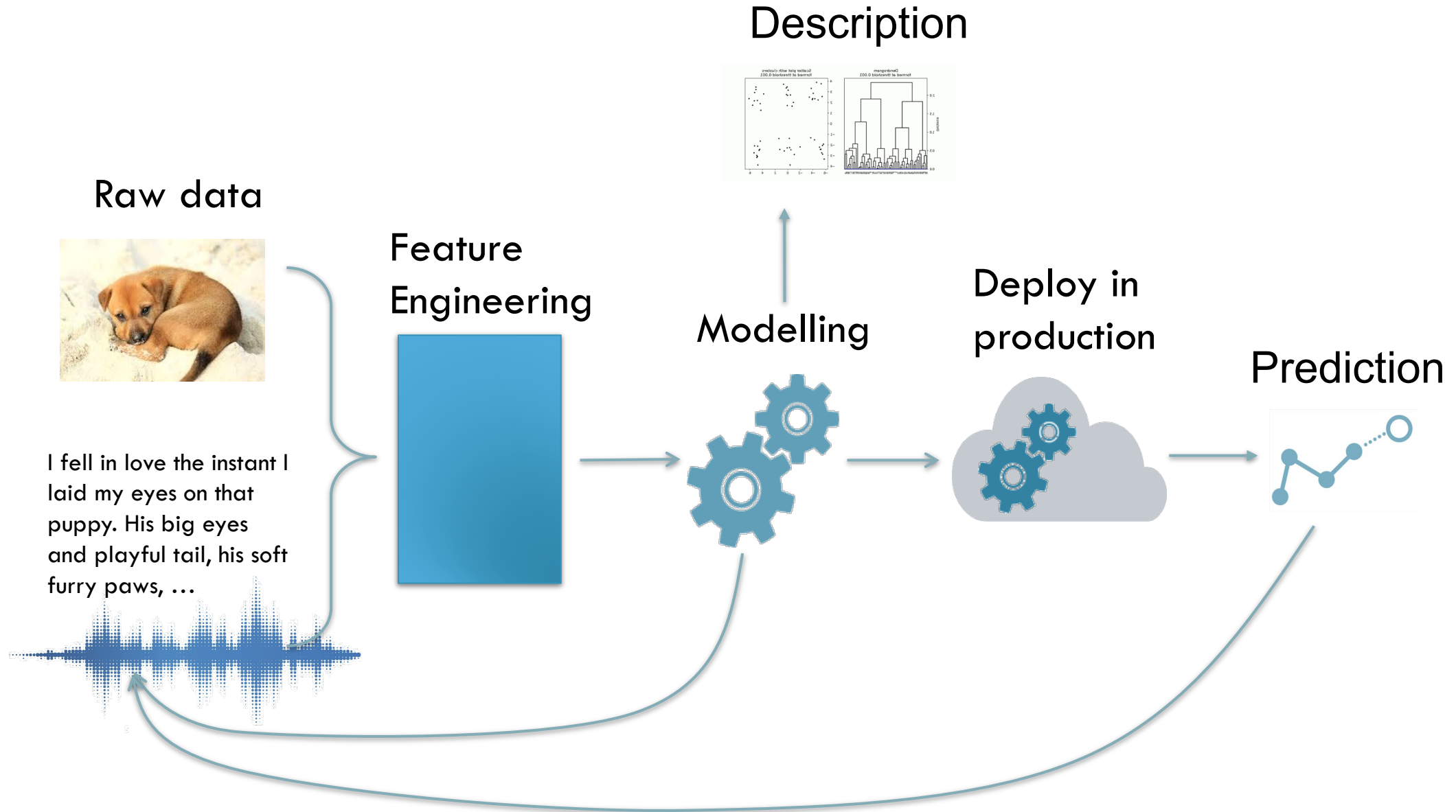
- Learning from the Trials and Errors

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Machine Learning Pipeline



ML Workflow





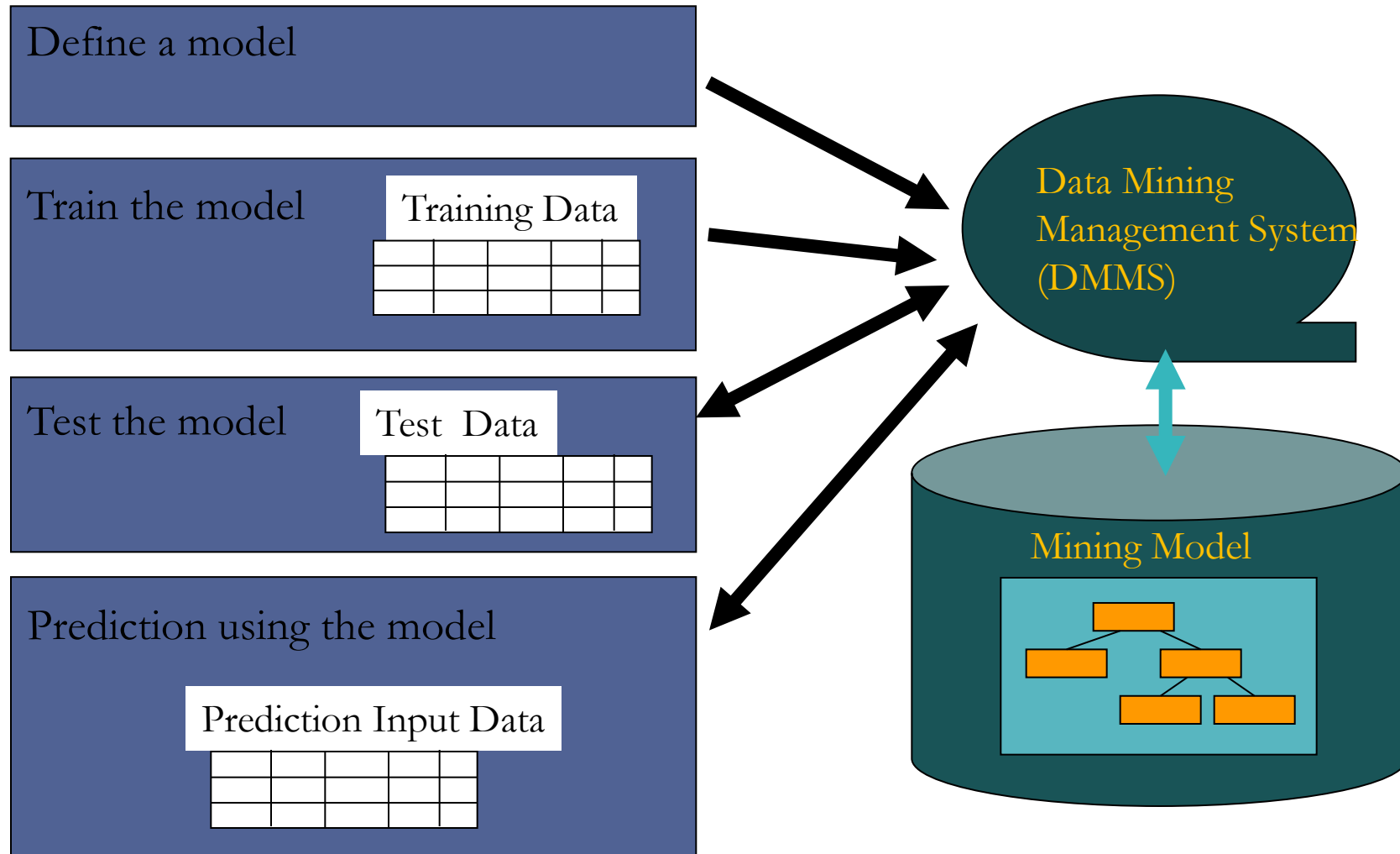
ML Workflow

ML workflow sketch:

1. Should I use ML on this problem?
 - ▶ Is there a pattern to detect?
 - ▶ Can I solve it analytically?
 - ▶ Do I have data?
2. Gather and organise data.
 - ▶ Preprocessing, cleaning, visualising.
3. Establishing a baseline.
4. Choosing a model, loss, regularisation, ...
5. Optimisation
6. Hyperparameter search.
7. Analyze performance & mistakes and iterate back to previous steps if needed.



A Typical Supervised Learning Process



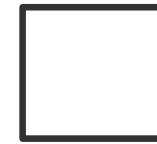


An instance or example

Supervised Learning

3	12	32	50	7	2	53	32	21	65	43	39	27	6	5	19	12
---	----	----	----	---	---	----	----	----	----	----	----	----	---	---	----	----

$\mathcal{X}$ = data/features for a subject



y = associated label 0/1

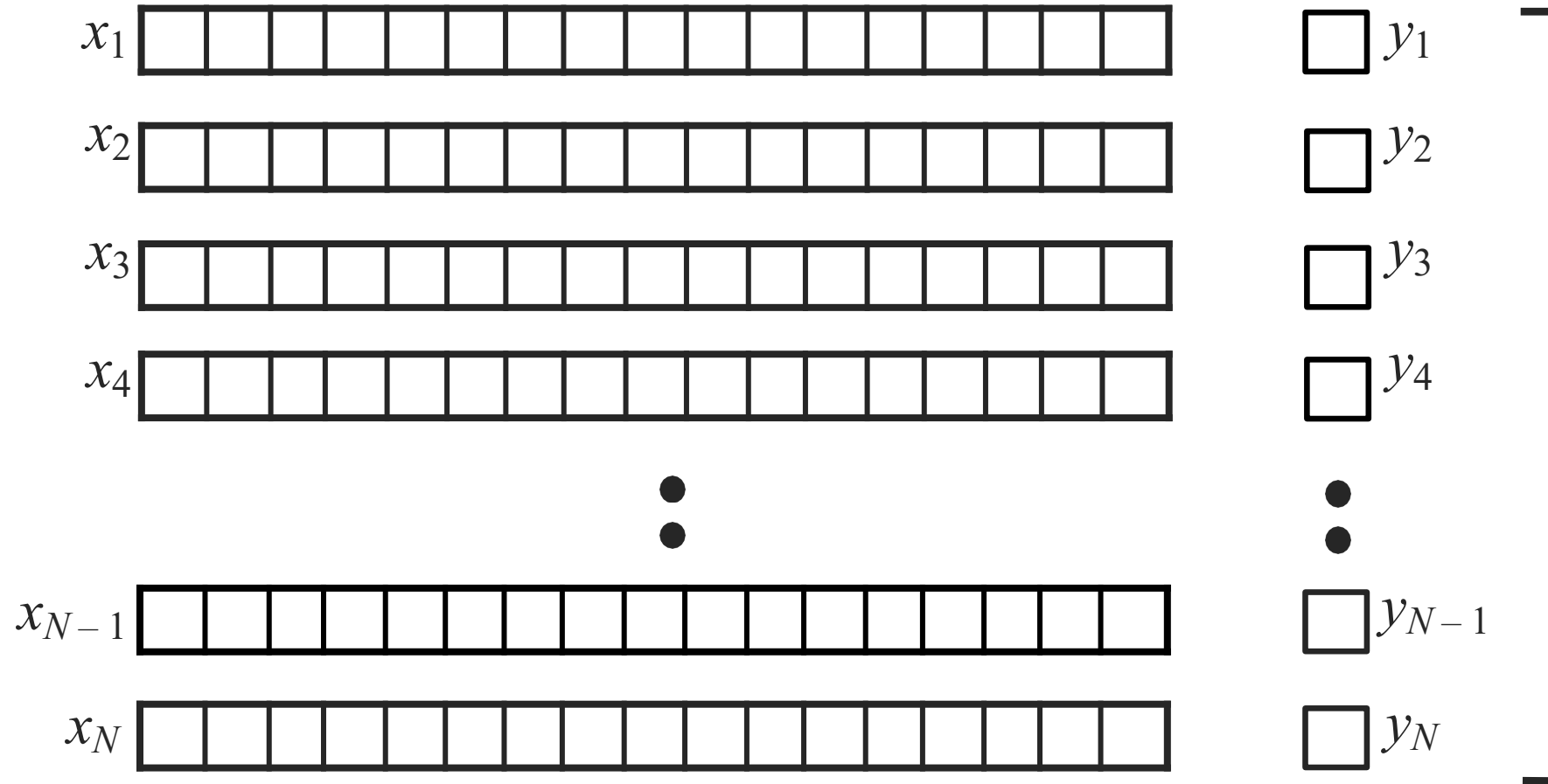
Unsupervised Learning

3	12	32	50	7	2	53	32	21	65	43	39	27	6	5	19	12
---	----	----	----	---	---	----	----	----	----	----	----	----	---	---	----	----

$\mathcal{X}$ = data/features for a subject

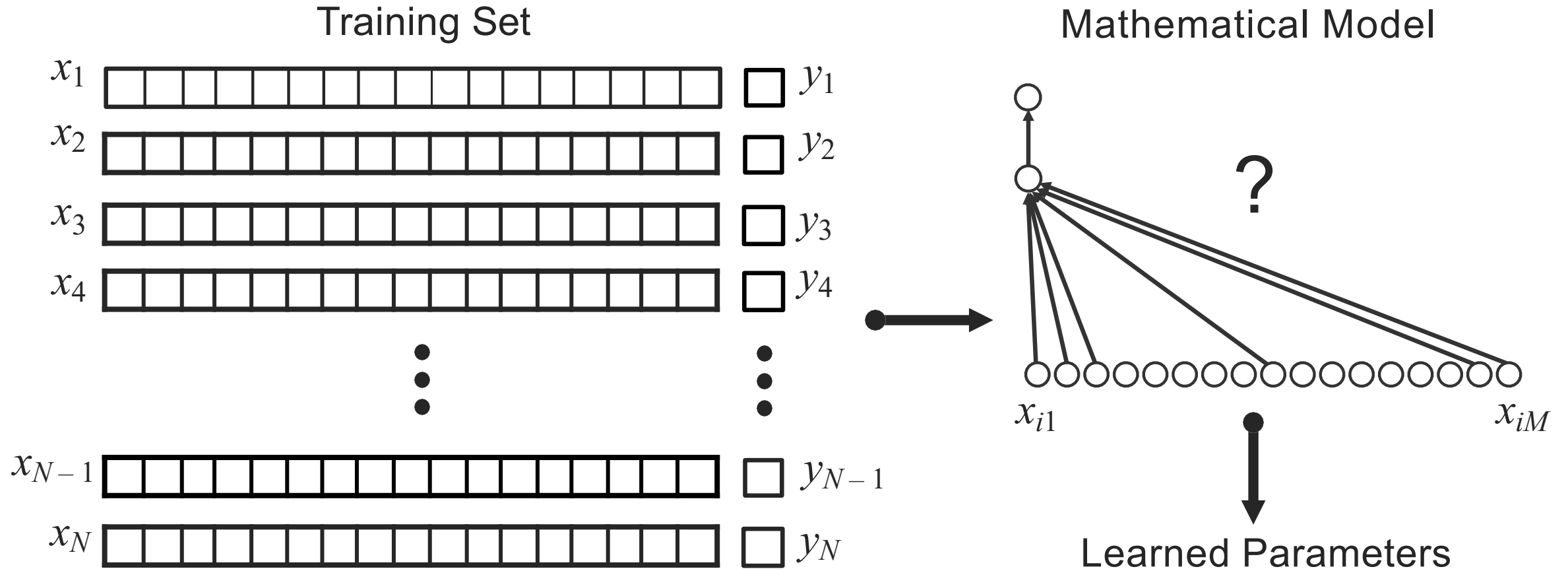


Training Set - Supervised Learning





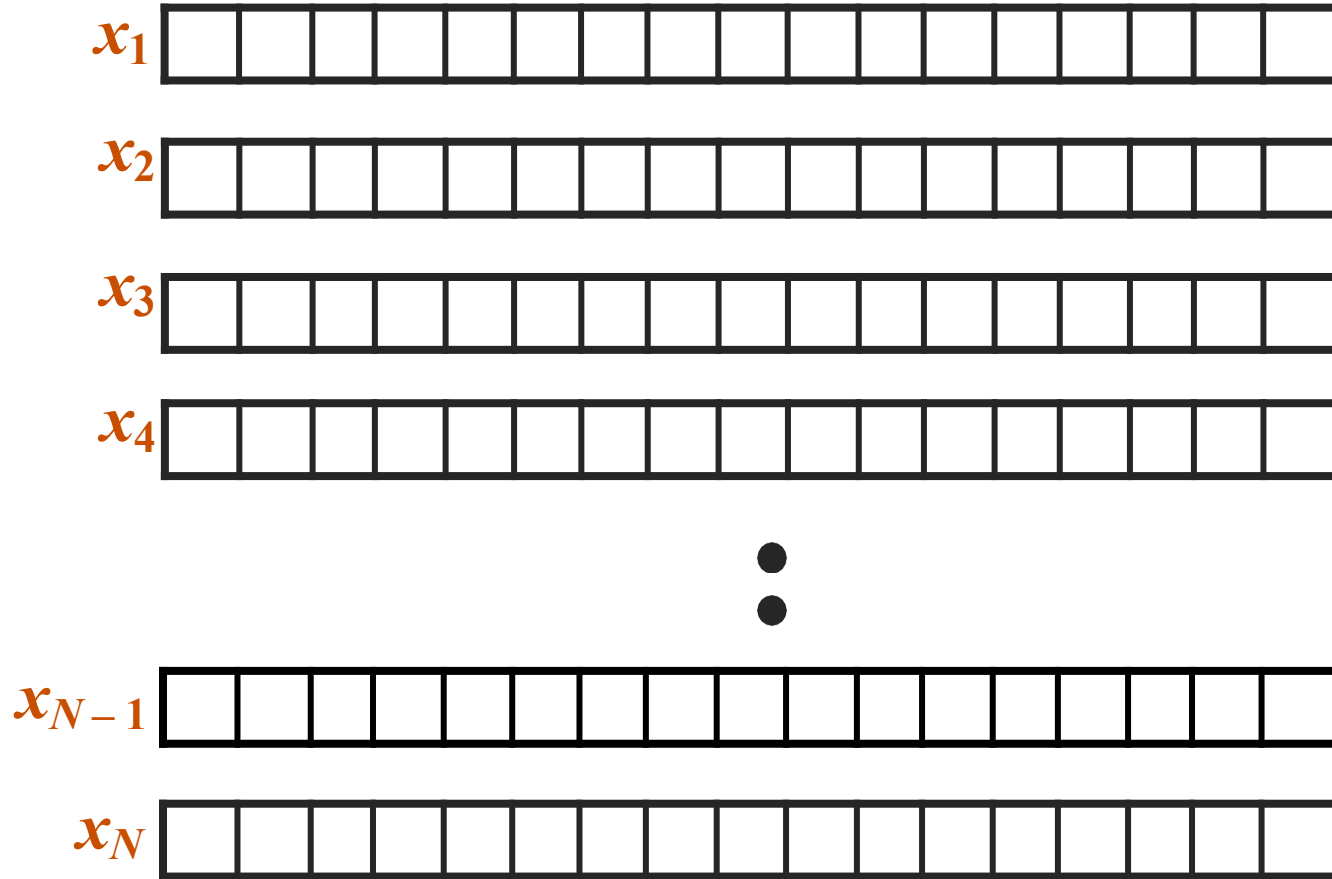
Learned Model Parameters – Supervised Learning





Training Set - Unsupervised Learning

$x = \text{data}$





Real-world Data Summary

- Unstructured (vs structured)
- Large scale
- Heterogeneous
- Big Data

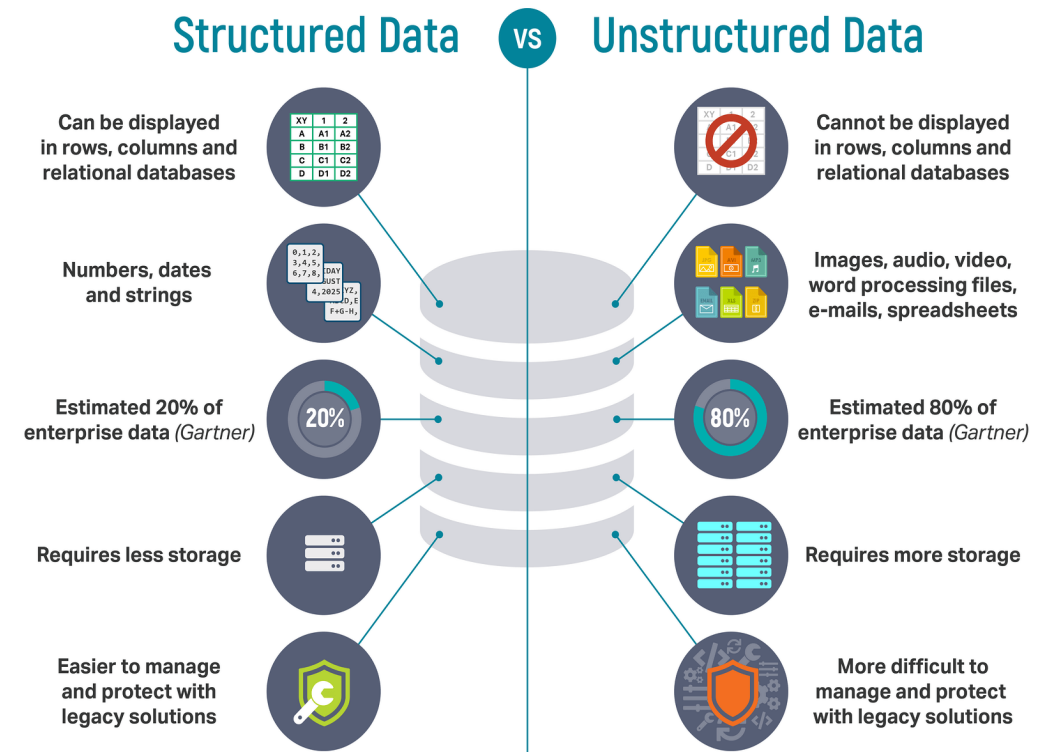


Image source: <https://lawtomated.com/structured-data-vs-unstructured-data-what-are-they-and-why-care/>



Machine Learning: Summary

- Machine learning discovers interesting patterns from a large amount of data.
- It is a complex process;
 - from data acquisition to data cleaning and pre-processing.
 - from data modelling to model deployment to knowledge dissemination.
- Several machine learning algorithms and models are available depending upon the output operation/task.
- Machine learning can be performed in a variety of information repositories.